An Edge Oriented Architecture for State Based Remaining Useful Life Prediction in Hydraulic Systems

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Abstract

Predictive maintenance in hydraulic systems is constrained in practice by the absence of run-to-failure data, limited component-level observability, and disconnected data pipelines between PLC-level operational events and machine-learning-based prognostics. This paper presents an edge-oriented prototype architecture that integrates these elements into a prototype-level state-based predictive maintenance workflow for hydraulic and fluid power systems.

The central methodological contribution is a formal state-transition Remaining Useful Life (RUL) formulation, which defines RUL as the remaining time, or number of hydraulic cycles, before the system transitions from nominal to PLC-supported abnormal or critical operation. This definition provides an operational

prognostic target without requiring complete run-to-failure trajectories. Supporting this formulation, an OPC UA-based acquisition and ETL framework aligns high-frequency hydraulic sensor data, multi-resolution storage, and PLC-based event logs into a structured data backbone for the investigated prototype.
The framework is evaluated on a real PLC-controlled electro-hydraulic prototype operated across 15 repeated sessions at 100 ms resolution. Latent operating states are inferred using K-Means and Hidden Markov Models and evaluated against PLC-based operational reference logs. K-Means achieved an accuracy of 0.737 and a binary anomaly F1-score of 0.938, while the HMM achieved an accuracy of 0.749, a macro F1-score of 0.635, and a binary anomaly F1-score of 0.933. For RUL estimation, XGBoost achieved a test MAE of 3.916 min and a median prognostic horizon of 34.068 min for the first abnormal transition ($\boldsymbol{R^2 = 0.911}$), while LSTM-Attention achieved the best critical RUL performance with a test MAE of 3.467 min, a median prognostic horizon of 39.662 min, and $\boldsymbol{R^2 = 0.964}$. SHAP, PDP, and attention-based analyses indicated that the models relied mainly on hydraulically plausible variables, including cylinder velocity, pressure ratios, oil flow rate, and cycle-level descriptors, supporting physical consistency without claiming causal inference.
The results provide prototype-level evidence that structured event logging, unsupervised state modeling, and state-transition RUL estimation can be integrated into a coherent hydraulic PdM architecture under practically relevant PLC-integrated constraints, including the absence of run-to-failure labels and restricted component-level observability.

Keywords: Predictive Maintenance, Intelligent Diagnosis, hydraulic systems, Remaining Useful Life (RUL), Long Short-Term Memory (LSTM),Hidden Markov Models (HMM).

Table 1: List of nomenclature

Nomenclature	Description
AI	Artificial Intelligence
ACR	Asset Criticality Ranking
CPS	Cyber-Physical System
ctrlX CORE	Programmable Logic Controller from Bosch Rexroth
DL	Deep Learning
Docker	Platform for developing, shipping, and running containerized applications
Dagster	Data orchestration platform for building and managing data pipelines
ERP	Enterprise Resource Planning
ETL	Extract, Transform, Load
FMECA	Failure Mode, Effects, and Criticality Analysis
GUI	Graphical User Interface
HMM	Hidden Markov Model
Industry 4.0	Fourth Industrial Revolution
LSTM	Long Short-Term Memory
MES	Manufacturing Execution System
ML	Machine Learning
OPC UA	Open Platform Communications Unified Architecture
PdM	Predictive Maintenance
PDP	Partial Dependence Plot (global marginal effect visualization method)
PLC	Programmable Logic Controller
PHM	Prognostics and Health Management
PH	Prognostic Horizon
PostgreSQL	Object-relational database management system
Power BI	Business Intelligence tool from Microsoft
ROI	Return On Investment
RUL	Remaining Useful Life
SAP	Systems, Applications, and Products in Data Processing
SAP PM	SAP Plant Maintenance
SHAP	SHapley Additive exPlanations
Solace	Middleware platform for event-driven streaming and message brokering
XGBoost	Extreme Gradient Boosting algorithm

1 Introduction

The Fourth Industrial Revolution, as presented by Lasi et al. [1], describes the IT-driven transformation of manufacturing systems, involving both technological and organizational innovation. Within this context, CPS enable new paradigms in production, where PdM plays a pivotal role. PdM relies on continuous monitoring of system conditions through sensor networks, which capture real-time operational data for subsequent analysis.

According to Zonta et al. [2], PdM harnesses historical records, domain knowledge, and computational models to forecast the behavior of critical assets. By applying statistical and machine learning methods, it identifies trends and anomalies that may indicate early stages of degradation or failure. This predictive capability is commonly intended to support proactive decision-making and reduce unplanned downtime.

As highlighted in [3], establishing an ACR is essential for identifying components that justify further FMECA. Once identified, these critical assets are subject to more advanced diagnostic and predictive strategies, optimizing the allocation of maintenance resources.

Dalzochio et al. [4] further emphasize that the deployment of ML-based PdM solutions in industrial environments faces core challenges related to data volume, latency, scalability, and bandwidth constraints. While edge computing is often proposed as a mitigation strategy, PdM does not inherently require real-time model execution. Instead, periodic analysis based on aggregated data collected over defined intervals is often sufficient to detect early signs of degradation [5].

As Pérez et al. [6] point out, critical events such as sudden failures or fast deviations from setpoints are typically addressed by built-in threshold-based alarms at the PLC level, which operate on millisecond timescales. These mechanisms provide immediate warnings to ensure process integrity and safety. In contrast, components undergoing slow degradation—often undetectable by standard alarms—are more effectively assessed using ML and DL models trained on longer time horizons. This functional separation supports a two-tiered monitoring approach: real-time visibility for safety and periodic analytics for predictive insights.

While state classification indicates the current operating condition, RUL estimation adds a temporal dimension by estimating how long the system can remain in an acceptable operating state before reaching an abnormal or critical condition. This time-to-transition information is more directly usable for maintenance planning, because it supports earlier intervention, prioritization of maintenance actions, and availability-oriented decision-making beyond instantaneous fault detection.

Contribution of this work. This paper presents an edge-oriented data and event architecture for state-based predictive maintenance in hydraulic and fluid power systems. The contribution is primarily architectural and methodological, and is validated at prototype level on a real PLC-controlled electro-hydraulic system. The work does not claim a universally transferable prognostic model or component-level causal diagnosis. Instead, it addresses a practical hydraulic monitoring setting in which run-to-failure data are unavailable, component-level observability is limited, and PLC-level operational events must be connected with machine-learning-based prognostics.

The work addresses four specific challenges in hydraulic predictive maintenance: (i) incomplete failure labels, (ii) restricted observability of component-level degradation, (iii) fragmented data pipelines between PLC events and analytical models, and (iv) the need for reproducible RUL targets under controlled but non-run-to-failure operating conditions.

The main contributions are as follows:

- **Edge-oriented data and event architecture for hydraulic systems.** An OPC UA-based acquisition and ETL framework is proposed to align high-frequency hydraulic sensor data, multi-resolution storage, and PLC-based operational event logs. The architecture provides a reproducible prototype-level data backbone for relating pressure–flow–motion measurements to operationally defined non-nominal machine states.
- **State-based hydraulic condition modeling under limited labels.** K-Means and Hidden Markov Models are used to infer latent operating states from unlabeled hydraulic time-series data. The resulting states are interpreted using hydraulic signal behavior, PLC event intervals, and expert knowledge, but they are not treated as physical ground-truth failure labels.

- **Formal state-transition RUL formulation.** RUL is defined as the remaining time, or number of hydraulic cycles, before the system transitions from nominal operation to a PLC-supported abnormal or critical operating state. This provides an operational prognostic target for hydraulic systems where complete run-to-failure trajectories are not available.
- **Quantitative prototype-level validation.** The framework is evaluated on 15 repeated hydraulic sessions acquired at 100 ms resolution. State identification is quantified using PLC-referenced classification metrics, while RUL prediction is evaluated using MAE, RMSE, R^2, an asymmetric PHM score and prognostic-horizon metrics. XGBoost and LSTM-Attention models are compared under target-aware temporal splits designed to reduce information leakage.
- **Explainability as physical consistency analysis.** SHAP values, PDPs, and attention-based analyses are used to evaluate whether the trained models rely on hydraulically plausible variables such as cylinder velocity, pressure ratios, oil flow rate, and cycle-level descriptors. These analyses support physical consistency, but they are not interpreted as proof of causal degradation mechanisms.

Overall, the contribution lies in integrating these elements into a prototype-level hydraulic PdM architecture with a formal state-transition RUL target and quantitative validation on the investigated system.

Paper organization. The remainder of this paper is organized as follows. Section 2 reviews related work on hydraulic predictive maintenance, data-driven fault diagnosis, RUL estimation, edge-oriented industrial architectures, and graph-based extensions for explainable prognostics. Section 3 describes the investigated electro-hydraulic prototype, including its physical configuration, failure modes, and PLC-oriented control abstraction. Section 4 presents the proposed edge-oriented data and event architecture, including PLC-based event logging, database design, and the ETL pipeline for multi-resolution sensor acquisition. Section 5 introduces the methodological contribution of the paper, namely the formal state-transition RUL formulation and its positioning with respect to existing RUL definitions. Section 6 describes the experimental dataset and the machine learning protocol, including data cleaning, feature engineering, unsupervised state modeling, temporal splitting, RUL model training, and explainability analysis. Section 7 reports the experimental results for state classification, XGBoost-based RUL regression, LSTM-Attention-based RUL estimation, and model explainability. Section 8 discusses the implications, limitations, observability constraints, and possible extensions toward physics-informed health-indicator RUL. Finally, Section 9 summarizes the main findings and outlines future research directions.

2 Related Work

The escalating costs associated with unplanned downtime underscore the need for proactive maintenance strategies that can both (i) detect incipient degradation and (ii) support actionable decision-making in realistic industrial settings. In this context, Prognostics and Health Management (PHM) aims to maximize RUL while mitigating unforeseen failures. However, practical deployment remains constrained by data availability, heterogeneous acquisition layers, multirate sensing, the need for architectural guidance toward scalable and interoperable integration across industrial IT/OT boundaries [7].

2.1 Electro-hydraulic modeling and control foundations

Electro-hydraulic actuators are strongly nonlinear systems exhibiting coupled pressure–flow–motion dynamics, motivating physics-based modeling as a reference for control, monitoring, and diagnostic abstraction. Representative studies derive continuous-time formulations using force balance, chamber continuity, and orifice-based valve flow relations, and validate them under realistic operating conditions. Sahu *et al.* characterize the static and dynamic behavior of a high-performance electro-hydraulic actuator and assess closed-loop performance for industrial suitability [8]. Xin *et al.* establish valve-controlled mathematical models for differential cylinders, deriving transfer functions under elastic and non-elastic loads, thereby providing a theoretical basis for analyzing static and dynamic characteristics [9]. From an implementation-oriented perspective, Adnan *et al.* demonstrate system identification from open-loop experimental data and real-time controller design via pole placement, highlighting the practicality of data-driven identification when full physical parametrization is infeasible [10]. Robustness considerations are emphasized by Niksefat and Sepehri, who design a force controller resilient to structured plant and environmental uncertainties using Quantitative Feedback Theory (QFT), illustrating the importance of uncertainty-aware designs in real deployments [11]. Collectively, these works motivate the PLC-oriented abstraction adopted in this paper, where discrete valve commands excite an underlying continuous plant and measured signals are mapped consistently to model variables for monitoring and learning.

Related integrated formulations can also be found in electro-hydraulic actuator (EHA) concepts, where motor, pump, and actuator dynamics are jointly represented within a unified hydrostatic model, supporting compact abstractions of coupled pressure–flow–motion behavior for monitoring purposes [12]. Recent simulation-based investigations further analyze circuit architectures and controller strategies for electro-hydrostatic actuators in mobile machinery, highlighting system-level performance and control-oriented design aspects [13, 14].

2.2 Data acquisition, synchronization, and industrial interoperability

While algorithmic advances in PHM are substantial, many studies under-specify how analysis-ready industrial datasets are acquired, synchronized, and maintained at scale. Nentwich and Reinhart highlight that PdM performance is tightly coupled to acquisition strategy design, proposing a systematic methodology to select monitored components, excitation trajectories, and sampling frequencies [15]. For hydraulic machinery, Qiu *et al.* explicitly address synchronization, anti-interference, and deployment complexity, proposing a fault-diagnosis-oriented acquisition system validated in power stations and pump test platforms [16]. Recent hardware-level contributions further emphasize scalable and synchronized measurement infrastructures: Toscani *et al.* propose a daisy-chain digital bus architecture supporting heterogeneous transducers with guaranteed synchronization for condition monitoring [17]. Beyond signal acquisition, interoperability of analytics models across manufacturing layers remains a critical barrier. Shin proposes an OPC UA-compliant interface for exchanging predictive models (PMML) via standardized information models, enabling publish/subscribe semantics for regression and neural-network models and fostering reusable industrial intelligence [18]. Ontology-driven integration approaches further address semantic consistency in cyber-physical production systems: Nuñez and Borsato introduce an ontology-based PHM model aligned with CPS architectures [19], while Brecher *et al.* demonstrate ontology-based data management for adaptable safety functions in distributed production environments [20]. These works collectively reinforce the architectural focus of the present paper, where OPC UA-based acquisition and edge-oriented ETL are treated as first-class design elements rather than implementation details.

2.3 Data-driven PdM: learning from data correlations and practical limitations

Data-driven predictive maintenance methods are grounded in machine learning, within which deep learning constitutes a specialized subset rather than an independent paradigm. Classical machine learning refers to feature-based models such as linear regression, support vector machines, decision trees, and ensemble methods, which operate on explicitly engineered descriptors.

Comprehensive surveys report a methodological shift within machine learning, from classical feature-driven approaches towards deep learning models capable of automated representation learning as data volumes and system complexity increase [21]. A structured overview of these learning paradigms, model families, and their inductive biases is provided in Table 3.

Despite their expressive capacity, data-driven models fundamentally learn statistical correlations present in the available data. Their predictive performance is therefore inherently constrained by the representativeness of the observed operating conditions and by whether physically meaningful degradation patterns are reflected in the data distribution. When relevant physical relationships are weak, masked by noise, or absent from the measured signals, purely data-driven approaches cannot recover them without additional structural information.

In realistic industrial environments, sensor data are multidimensional, noisy, and often nonstationary. Prior work highlights the importance of robust feature construction through trend extraction, unsupervised selection, and signal decomposition before prognostic modeling [22]. Such considerations underline the central role of preprocessing and feature engineering in data-driven PdM.

For contextualization, Table 2 summarizes commonly adopted physics-based, data-driven, and hybrid modeling paradigms in condition monitoring and predictive maintenance.

Table 2: Taxonomy of modeling approaches commonly adopted in condition monitoring and predictive maintenance, synthesized from recent surveys on physics-based, data-driven, and physics-informed machine learning models [23, 24].

Category	**Typical formulations / examples**
Physics-Based Models	First-principles models; analytical / white-box models; simulation-based models (Finite Element Method (FEM) / Computational Fluid Dynamics (CFD)).
Data-Driven Models	Machine learning: classical machine learning (ML); deep learning including Multilayer Perceptrons (MLP), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Transformers, and Graph Neural Networks (GNN).
Hybrid / Physics-Informed Models	Grey-box models; Physics-Informed Neural Networks (PINNs); physics-guided deep learning; causality- or physics-guided Graph Neural Networks (GNNs).

Table 3: Machine learning paradigms and representative model families, summarizing commonly used architectures, inductive biases, and data modalities as reported in recent surveys on deep learning, predictive maintenance, multi-objective learning, and graph-based spatio-temporal modeling [24–27].

Learning paradigm	**Model family / representative examples**	**Inductive bias (summary)**	**Typical data**
Supervised	**Classical ML:** linear and logistic regression; Support Vector Machines (SVM); k-Nearest Neighbors (kNN); decision trees; random forests; gradient boosting	Linearity; margin maximization; locality in feature space; rule-based partitioning; ensemble averaging	Labeled tabular features; engineered signals
	Deep Learning: Multilayer Perceptrons (MLP); Convolutional Neural Networks (CNN); Recurrent Neural Networks (RNN), including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU); Transformers (including encoder–decoder architectures)	Representation learning; spatial locality (CNN); temporal order (RNN); global attention mechanisms (Transformers)	Images; time series; sequences
	Graph learning: Graph Neural Networks (GNNs) and graph transformers	Relational and topological inductive bias (message passing or attention over graphs)	Graph-structured systems; networks
Unsupervised	**Classical ML:** k-means clustering; hierarchical clustering; Principal Component Analysis (PCA)	Cluster geometry assumptions; nested partitions; linear subspace representations	Unlabeled features; embeddings
	Deep representation / generative learning: autoencoders (sparse and denoising); Variational Autoencoders (VAE); diffusion models	Compressible latent representations; reconstruction-based learning; probabilistic latent spaces; iterative denoising generative priors	High-dimensional signals; anomaly detection; generative modeling
	Graph learning: unsupervised or generative graph models (via autoencoders, contrastive learning, or diffusion)	Relational representations without labels; structural priors derived from graph topology	Unlabeled graphs; topology with measurements
Self-supervised	**Representation learning:** contrastive learning (e.g., Simple Framework for Contrastive Learning of Visual Representations (SimCLR), Momentum Contrast (MoCo)); masked prediction (e.g., Masked Autoencoders (MAE), Masked Language Modeling (MLM)); self-distillation	Learning invariances and context completion via pretext tasks; scalable representation learning	Large unlabeled datasets (images, sensor signals, text)
	Foundation models: Transformers and Large Language Models (LLMs) (pretraining followed by task-specific adaptation)	Attention-based sequence modeling; broad statistical priors acquired through large-scale pretraining	Text and code (LLMs); multimodal corpora; tokenized signals
	Graph self-supervised learning: contrastive or masked objectives on graphs; graph transformers	Topology-aware relational representation learning without explicit labels	Large graph datasets; relational sensor graphs
Reinforcement learning	**Classical RL:** Q-learning; State–Action–Reward–State–Action (SARSA)	Markov Decision Process (MDP) assumption; value-based learning in discrete state–action spaces	Interactive environments; low-dimensional control
	Deep RL: Deep Q-Networks (DQN); policy gradient methods; actor–critic methods	Neural function approximation; exploration versus exploitation; learned state representations	High-dimensional observations; robotics and control
	RL for foundation models: preference optimization and Reinforcement Learning from Human Feedback (RLHF) pipelines	Reward modeling and alignment via preferences; policy optimization in output space	Human feedback; instruction data

2.4 Hydraulic fault diagnosis under multirate sampling

Hydraulic systems typically involve heterogeneous sensors operating at different sampling rates, complicating temporal alignment and end-to-end learning. Huang *et al.* propose a deep learning framework explicitly designed for multirate hydraulic data, achieving robust diagnostic accuracy even under severe sampling imbalance and class skew [28]. Complementary multirate diagnostic architectures integrate efficient convolution variants and feature fusion mechanisms: Qu *et al.* combine residual shrinkage separable CNNs with statistical features and gated attention fusion to aggregate high- and low-rate information [29], while Zhang *et al.* employ fully convolutional networks for multi-fault diagnosis from multirate time series [30]. Beyond deep learning, Zheng *et al.* introduce a total multirate linear Gaussian state-space model that separates global dynamic latent variables from rate-specific static components, enabling statistically grounded fault detection across sampling regimes [31]. These contributions motivate the multi-resolution data backbone adopted in this paper, combining high-frequency edge acquisition with aggregated analytical views suitable for scalable learning.

From an industrial perspective, recent experimental evidence highlights that hydraulic degradation is tightly linked to specific operating regimes rather than isolated fault events. In particular, load-holding operation has been shown to accelerate wear progression in axial piston pumps, resulting in increased internal leakage and altered pressure dynamics, which can serve as reliable health indicators for data-driven prognostics [32].

2.5 RUL estimation under limited and heterogeneous degradation data

RUL prediction remains challenging due to scarce run-to-failure datasets and heterogeneous operating conditions. Health-index-based approaches provide a practical abstraction: Liu *et al.* propose RUL estimation via health index similarity using distance and directional metrics [33], while Yang *et al.* introduce dynamic smoothing and ensemble realization to capture nonlinear degradation trends [34]. To address domain shift and incomplete degradation trajectories, Li *et al.* propose partial domain adaptation using adversarial learning when only early target-domain degradation data are available [35], and Cheng *et al.* combine dynamic modeling with transfer learning and generative data augmentation to mitigate the lack of full-cycle degradation data [36]. More recent work reframes prognostics by modeling degradation trajectories directly: Fan *et al.* propose Degradation Path Approximation (DPA), deriving RUL from aligned health indicator trajectories under limited historical data [37]. However, such formulations typically assume continuous and monotonic degradation, whereas industrial electro-hydraulic systems often exhibit regime changes and economically non-viable operating states rather than smooth failure trajectories. This motivates alternative RUL formulations grounded in state transitions and operational thresholds, as adopted in the present work.

2.6 Open challenges in hydraulic predictive maintenance

Although predictive maintenance challenges such as limited data availability, restricted explainability, and poor cross-system transferability are widely reported across industrial domains, the literature provides comparatively fewer explicit syntheses focused on hydraulic systems. Reviews in adjacent predictive maintenance fields consistently identify data quality, model interpretability, scalability, and generalization as persistent barriers to industrial deployment [38–41]. These challenges are also relevant in hydraulic applications, but they require more careful interpretation because hydraulic behavior is strongly dependent on circuit topology, sensing configuration, and operating regime.

Acquisition strategy and observability. A first challenge concerns the acquisition of sufficiently informative temporal data. In hydraulic systems, degradation-relevant effects such as leakage-induced motion irregularities, short pressure–flow transients, and cycle-to-cycle performance variations may be attenuated or lost when measurements are observed only at coarse temporal resolutions. Accordingly, predictive maintenance performance depends strongly on acquisition strategy, synchronization quality, and the match between sensing resolution and the underlying hydraulic dynamics [15, 16].

Inter-system variability and limited transferability. A second challenge is the structural variability of hydraulic circuits. Even nominally similar systems may differ in valve configuration, control logic, actuator type, load conditions, and sensor placement, producing substantially different pressure–flow–motion relationships. As a result, models trained on one hydraulic installation cannot be assumed transferable to another without adaptation. Under these conditions, monitoring strategies must focus on the most observable and operationally relevant subsystems, in particular the power unit and the actuator, where degradation becomes visible through efficiency loss, slowdown, or motion anomalies [38, 42].

Explainability under incomplete observability. A third open issue is the physical interpretation of data-driven predictions. Methods such as SHAP, PDPs, or attention maps can identify statistically influential variables, but they do not by themselves establish causal hydraulic relationships. Their usefulness therefore remains bounded by sensor coverage and by whether physically meaningful degradation patterns are sufficiently encoded in the measured data. Consequently, explainability in hydraulic PdM should be interpreted as evidence of consistency with observed system behavior rather than as proof of physical causality, motivating future extensions based on causal graph modeling and physics-informed relational representations [39, 41, 42].

Graph-based and physics-informed extensions. Recent graph-based prognostic models have shown that Graph Neural Networks (GNNs) can improve fault diagnosis and RUL estimation by representing sensors, components, or extracted features as nodes and by modeling spatial–temporal dependencies through graph construction, graph modeling, information fusion, and graph-level readout [43, 44]. Such approaches are particularly relevant for hydraulic systems, where degradation is rarely expressed by a single signal and instead emerges from coupled pressure–flow–motion interactions among the power unit, valves, linear or rotary actuators, and load-dependent operating regimes. Recent causal graph frameworks, such as DCMLDF, extend this idea by combining multi-scale temporal encoding, adaptive causal graph inference, semantics-aware graph aggregation, and a causality-guided predictive decoder to model dynamic fault evolution from multivariate sensor data [45].

However, graph-based explanations must be interpreted cautiously, because many data-driven graph structures are inferred from statistical similarity, correlation, or learned attention weights; therefore, graph edges do not necessarily represent physical causality or mechanism-level degradation relationships. Physics-informed graph modeling can reduce this limitation by constraining nodes, edges, or message passing with prior knowledge about hydraulic topology, flow direction, pressure coupling, actuator kinematics, physical constraints, and degradation mechanisms [23, 44]. In the present work, graph-based causal learning is not used as the primary prognostic model because the available measurements mainly support system-level observability of the power unit and actuator, rather than full component-level causal identification. Instead, GNN- and DCMLDF-type models are discussed as future extensions toward physics-informed health-indicator RUL once richer observability becomes available at selected hydraulic nodes, such as the pump or power unit, valve stage, actuator chambers, leakage path, temperature path, and linear or rotary actuator output.

RUL definition in hydraulic contexts. Finally, the definition of Remaining Useful Life remains particularly challenging in hydraulic systems. Conventional prognostic formulations often assume a continuous and monotonic degradation trajectory toward physical failure. In many hydraulic applications, however, maintenance decisions are triggered earlier, when actuator-level performance degradation produces unacceptable operational or economic consequences. RUL is therefore often more meaningfully defined with respect to operationally unacceptable performance states than to catastrophic failure alone, which motivates state-based prognostic formulations in hydraulic applications [32, 42].

2.7 Architectural perspectives: streaming, data management, and scalability

Despite algorithmic maturity, scalable PHM deployment critically depends on data ingestion, streaming, and storage architectures. Kappa-style streaming architectures have been shown to reduce latency and complexity compared to hybrid Lambda designs in continuous real-time analytics [46, 47]. Scalable stream aggregation and fault-tolerant sensor analytics are further addressed by Henning and Hasselbring, who demonstrate reliable multidimensional aggregation in Industry 4.0 data streams [48]. At the persistence layer, comparative studies show that NoSQL and time-series databases are well suited for high-rate industrial data ingestion [49, 50], while relational and warehouse-oriented systems remain advantageous for structured metadata, cross-site reporting, and analytical traceability [51, 52]. Recent surveys on data ingestion pipelines and IIoT architectures further highlight the importance of edge–cloud partitioning, protocol interoperability, and scalable ETL design for industrial AI systems [53–55]. In the present work, scalability is therefore considered at the architectural-design level, not as an experimentally demonstrated fleet-level property. The prototype implementation evaluates whether the proposed data and event structure can support state-based PdM within one PLC-controlled hydraulic system.

2.8 Positioning of this work

Much of the existing literature on fault diagnosis and RUL estimation emphasizes increasingly sophisticated machine learning and deep learning models, often assuming the availability of degradation labels or health indicators that are consistent and physically meaningful. In practical hydraulic systems, this assumption rarely holds.

Hydraulic degradation typically manifests through discrete operational events, such as abnormal pressure regimes, temperature-induced slowdowns, or system downtime driven by economic or maintenance-related constraints, rather than as smooth numerical trajectories. Without a structured record of faults, stoppages, and maintenance activities, RUL predictions, regardless of model complexity, remain largely referential and fail to capture their full industrial applicability.

While several frameworks propose end-to-end predictive maintenance pipelines, data acquisition and event logging are frequently treated as secondary implementation aspects. As a result, the connection between model outputs and the underlying physical and operational causes of degradation remains weak, limiting both interpretability and economic value.

The present work is positioned on the premise that operationally interpretable RUL estimation can be supported when data-driven models are supported by sustainable data acquisition and structured records of operational events. Using a real-world electro-hydraulic case study controlled by an industrial PLC, this work demonstrates how fault and maintenance records provide the necessary context to interpret model outputs and assess the practical relevance of model outputs in the investigated hydraulic system.

3 Hydraulic Prototype System Description

This section describes the electro-hydraulic prototype used as the experimental testbed in this study. The setup emulates the cyclic extension–retraction operation of a hydraulic press and combines industrial hydraulic components, PLC-based control, and OPC UA-based data acquisition. The main components are listed in Table 4, and the hydraulic circuit is shown in Figure 1.

Table 4: Rexroth components list.

Item	Quantity	Designation	Type	Rexroth material num.
1.10	1	Power Unit	XITE Hydraulix 300 workstation	R961009456
2.40	1	Gear wheel flow rate sensor	TS-HC-GFM 3843-03-S-3	R961002508
1.30-31-32	3	Throttle valve	TS-HC-DV 06-1-1X/V	R961002539
1.40	1	Pressure relief valve	TS-HC-DBDH 6 G1X/100&	R961002520
1.60	1	4/3 directional control valve	TS-HC-4WE 6 G6X/EG24&	R961002548
1.70	1	Pressure sequence valve	TS-HC-DZ 6 DP1-5X/75Y	R961002558
1.80	1	4/2 directional control valve	TS-HC-4WE 6 C6X/EG24&	R961002547
1.90	1	Load unit, 80 kg (stationary)	TS-HC-CD 70H 25/16-&	R961004005
2.30-31-32	3	Pressure sensor	TS-HC-3403-17-S-E5.37	R901465414
2.50	1	Position sensor	–	–
-	3	Hydraulic distributor (4 ports)	TS-HC-4F-3M1W	R961002485

Note: Component data are taken from Bosch Rexroth specification sheets [56].

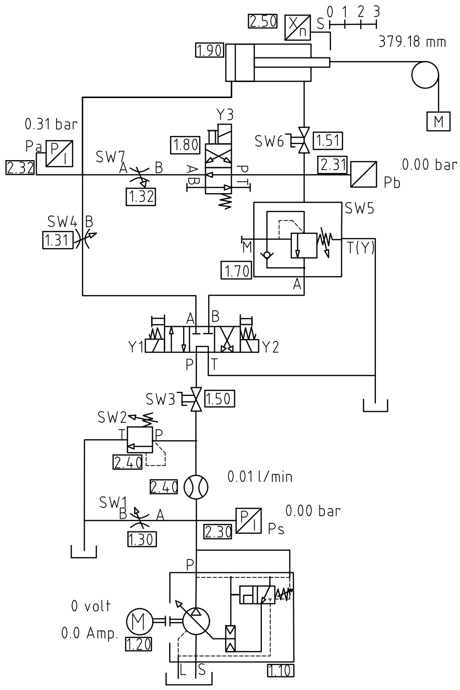

Fig. 1: Hydraulic drawing of the system.

The system is supervised by a PLC that executes standard threshold-based logic for operational safety. For this study, the PLC is interfaced with a ctrlX CORE platform. The integration of an OPC UA server on this controller is pivotal to the proposed dataflow framework, enabling the high-frequency acquisition of time-series sensor data. This data is subsequently stored in a structured database and serves as the input for the prediction pipelines developed for anomaly detection and prognostics.

For our analysis, we assume a continuous extension and retraction cycle of the cylinder. Within this operational context, we specifically investigate two controlled degradation scenarios:

- **Excessive Internal Leakage:** Leading to cylinder stoppage due to significant internal fluid loss.
- **Anomalous Operation (Slowdown):** Characterized by a sudden increase in oil temperature, causing a deceleration in cylinder extension and retraction.

The displayed hydraulic distribution allows for the simulation of internal cylinder leakage, specifically between the high and low-pressure zones, by activating a valve controlled by relay Y3 as shown in Figure 1. Critically, this system intentionally does not include a cooling mechanism. Consequently, over time, the oil temperature gradually increases, leading to anomalous operating conditions. This situation is further exacerbated when internal cylinder leakage is simulated, often culminating in the loss of repeatable cylinder motion under critical operating conditions.

This foundational setup, established through the construction of fault detection tables (Section 4.1), enabled the identification of potential system failures. Moreover, it defined the forward path for implementing predictive models specifically aimed at detecting and forecasting two operational degradation patterns: excessive internal leakage and anomalous operation (slowdown). These models integrate the corresponding physical variables and incorporate insights from hydraulic experts.

For safety reasons in a controlled commissioning and training environment, and despite the use of real hydraulic system components and configuration, a cooling system was intentionally not installed. The primary objective of this setup is didactic and educational.

3.1 Mathematical Model and PLC-Oriented Control Abstraction

Table 5 summarizes the symbols, units, and signal representations used in the hydraulic-cylinder model and subsequent equations

The hydraulic actuator is controlled by a PLC through discrete valve commands, resulting in a hybrid continuous–discrete control structure. The PLC does not directly implement a continuous controller, but instead selects discrete valve states that excite the underlying continuous hydraulic dynamics. For this reason, the physical system is modeled as a continuous-time plant driven by a discrete-valued control input.

The plant dynamics follow well-established electro-hydraulic actuator formulations, including force balance, chamber continuity, and orifice-based valve flow relations [8, 9, 11].

The cylinder kinematics are given by

$$\dot{X}_{\mathrm{cyl}} = V_{\mathrm{cyl}}. \tag{1}$$

The translational dynamics follow from Newton's second law:

$$\begin{aligned} m\,\dot{V}_{\mathrm{cyl}} = {} & A_A P_A - A_B P_B \\ & - F_{\mathrm{fric}}(V_{\mathrm{cyl}}, T) - F_{\mathrm{load}}(X_{\mathrm{cyl}}, t). \end{aligned} \tag{2}$$

A compact friction representation is adopted:

$$F_{\mathrm{fric}}(V_{\mathrm{cyl}}, T) = F_c(T)\,\mathrm{sgn}(V_{\mathrm{cyl}}) + b(T)\,V_{\mathrm{cyl}}. \tag{3}$$

For consistency with the feature set used in data-driven analysis, the retraction velocity is defined as

$$V_{\mathrm{ret}} = \min(V_{\mathrm{cyl}}, 0)\,. \tag{4}$$

3.1.1 Hydraulic Pressure Dynamics

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Table 5: Nomenclature aligned with measured signals and the hydraulic-cylinder model

Symbol	**Unit**	**Meaning and relation to measured signals**
X_{cyl}	mm	Cylinder position (measured position sensor); model state in meters.
$V_{\text{cyl}} = \dot{X}_{\text{cyl}}$	mm/s	Cylinder velocity (measured); model state in meters per second.
V_{ret}	mm/s	Retraction velocity feature, e.g. $V_{\text{ret}} = \min(V_{\text{cyl}}, 0)$.
m	kg	Equivalent moving mass (piston, rod, and load).
P_A	bar	Pressure at port/chamber A (extension side).
P_B	bar	Pressure at port/chamber B (retraction side).
P_S	bar	Supply pressure (HPU / supply line).
P_T	bar	Tank/return pressure (often near constant).
A_A	m^2	Effective piston area, chamber A.
A_B	m^2	Effective piston area, chamber B.
$V_A(X_{\text{cyl}})$	m^3	Fluid volume in chamber A as a function of piston position.
$V_B(X_{\text{cyl}})$	m^3	Fluid volume in chamber B as a function of piston position.
Q_A	m^3/s	Flow into chamber A.
Q_B	m^3/s	Flow into chamber B.
Q_{oil}	l/min	Oil flow-rate signal; optional mapped feature from model flows.
$\beta_e(T)$	Pa	Effective bulk modulus (temperature dependent).
ρ	$\mathrm{kg/m}^3$	Hydraulic oil density.
$F_{\text{fric}}(V_{\text{cyl}}, T)$	N	Friction force (velocity and temperature dependent).
$F_{\text{load}}(X_{\text{cyl}}, t)$	N	External load force acting on the actuator.
$C_d(T)$	–	Valve discharge coefficient (temperature dependent).
$A_v(u)$	m^2	Effective valve orifice area as a function of PLC command u.
$C_\ell(T)$	$\mathrm{m}^3/\mathrm{s/Pa}$	Internal leakage coefficient (temperature dependent).
u	–	Discrete PLC valve command: $u \in \{-1, 0, +1\}$ (retract / stop / extend).
T	°C	Hydraulic oil temperature.
$P_{A/S}$	–	Pressure ratio feature $P_{A/S} = P_A/P_S$.
$P_{B/A}$	–	Pressure ratio feature $P_{B/A} = P_B/P_A$.

Note: The physical model is typically implemented in SI units (m, s, Pa). Measured signals may be stored in engineering units (mm, mm/s, bar, l/min); consistent unit conversion is applied during identification and machine-learning preprocessing.

The chamber pressure dynamics follow continuity:

$$\dot{P}_A = \frac{\beta_e(T)}{V_A(X_{\text{cyl}})}\Big(Q_A - A_A V_{\text{cyl}} - C_\ell(T)\,(P_A - P_B)\Big), \tag{5}$$

$$\dot{P}_B = \frac{\beta_e(T)}{V_B(X_{\text{cyl}})}\Big(Q_B + A_B V_{\text{cyl}} - C_\ell(T)\,(P_B - P_A)\Big). \tag{6}$$

The chamber volumes depend on piston position:

$$\begin{aligned} V_A(X_{\text{cyl}}) &= V_{A0} + A_A X_{\text{cyl}}, \\ V_B(X_{\text{cyl}}) &= V_{B0} + A_B\,(L - X_{\text{cyl}})\,. \end{aligned} \tag{7}$$

3.1.2 Valve Flow Model and PLC Interface

The directional valve is actuated by a PLC with discrete command

$$u \in \{-1,\, 0,\, +1\}, \tag{8}$$

where $u = +1$ denotes extension, $u = -1$ retraction, and $u = 0$ a blocked/hold state.

A minimal orifice-based mapping from discrete logic to continuous flows is

$$\begin{aligned} Q_A &= C_d(T)\, A_v(u) \sqrt{\frac{2}{\rho}} \\ &\quad \times \sqrt{\max(P_S - P_A,\, 0)}, \end{aligned} \tag{9}$$

$$\begin{aligned} Q_B &= -\; C_d(T)\, A_v(u) \sqrt{\frac{2}{\rho}} \\ &\quad \times \sqrt{\max(P_B - P_T,\, 0)}. \end{aligned} \tag{10}$$

To match the plotted features used for condition monitoring, we additionally define

$$P_{A/S} = \frac{P_A}{P_S}, \qquad P_{B/A} = \frac{P_B}{P_A}. \tag{11}$$

The flow-rate signal is reported as a derived channel:

$$Q_{\text{oil}} \triangleq \alpha\, |Q_A| \quad (\text{or } \alpha\, |Q_B|), \tag{12}$$

This notation ensures that (i) the physics-based states map directly to measured channels $(X_{\text{cyl}}, V_{\text{cyl}}, P_A, P_B, P_S)$ and (ii) the PLC-level discrete decision u is explicitly connected to the continuous hydraulic dynamics used to motivate the measured variables and ML feature construction.

Remark on PLC Signal Conditioning

Measured signals are subject to standard PLC scaling and low-pass filtering, which can be approximated by first-order (PT1) dynamics and are therefore not modeled explicitly.

3.2 Observability Scope and Accelerated Degradation Protocol

The experimental setup is not intended to reproduce the complete lifetime degradation of individual hydraulic components. Instead, it is used to evaluate whether the proposed edge-oriented data and event architecture can provide a basis for system-level condition monitoring and state-transition RUL estimation under controlled hydraulic degradation patterns. This scope reflects a common industrial constraint: component degradation such as pump wear, seal deterioration, valve wear, or viscosity-related efficiency loss usually evolves over long operating periods and requires long-term industrial data, verified maintenance records, and component replacement histories.

Accordingly, the monitoring focus is defined at system level and concentrated on the most observable hydraulic subsystems: the power unit and the actuator. The power unit is represented through supply pressure and oil-flow behavior, while the actuator is represented through cylinder position, velocity, pressure ratios, and cycle-level motion descriptors. To make degradation-relevant behavior observable within a feasible laboratory acquisition window, the prototype was operated without active cooling and with increased internal leakage through the controlled leakage path. The resulting data are therefore interpreted as accelerated system-level performance degradation, not as naturally accumulated long-term component wear. Future work may investigate whether this architecture can be extended toward component-level degradation analysis in power units and linear or rotary actuators by increasing observability, using longer industrial operation, verified maintenance records, and component-specific degradation labels.

4 Edge-Oriented Data and Event Architecture

This section presents the main architectural components used to support the proposed prototype-level predictive maintenance workflow:

- A database architecture for structured recording of machine faults and downtime events.
- A data flow architecture (ETL pipeline) to collect, process, and store real-time sensor data from PLCs into time-series tables.

Together, these architectures provide a unified data backbone for evaluating anomaly detection and RUL prediction models in the investigated prototype.

In industrial environments, anticipating equipment degradation provides critical lead time for proactive maintenance planning, enabling a transition from reactive interventions to scheduled and economically optimized actions. However, the practical deployment of prognostic models is often hindered by the scarcity of complete run-to-failure datasets, as catastrophic breakdowns are rare, safety-critical, and actively avoided in real systems.

To address this limitation, RUL is redefined in this work not as the time to physical component failure, but as the expected time until the system transitions into an operational state that is no longer acceptable according to PLC-supported warning or critical condition rules. These operational states are identified through a data-driven state model based on K-Means clustering and HMMs, and are validated using expert knowledge and structured downtime records.

Accordingly, RUL prediction is formulated as a state-transition forecasting problem, where the target variable represents the remaining time until a predefined PLC-supported abnormal or critical operating state is reached, rather than the extrapolation of a continuous degradation trajectory.

4.1 Event-Based Data Architecture for Fault Logging and Operational States

A minimal yet standardized relational schema is proposed to support the automatic recording of machine faults and downtimes in a structured and query efficient manner. The design enables both consistent taxonomy for supervised learning and traceable performance monitoring aligned with Industry 4.0 principles. When a non-operational state is detected—via the `run_status` signal derived from sensor thresholds or PLC logic—an event is automatically logged into the database. A graphical interface allows operators to validate and enrich these records with contextual details (e.g., fault codes, failure causes, or maintenance notes), addressing limitations of purely sensor-driven diagnosis.

Importantly, the proposed database is not designed as a repository of catastrophic failure events, which are typically scarce in industrial practice, but as a structured entry point for exploratory analysis. It captures equipment stoppages and non-operational states together with their operationally defined reasons, even when detailed fault mechanisms are not yet fully identified. This event-centric representation establishes the initial operational context required to relate time-series sensor data to system behavior, and provides a foundation for subsequent physical interpretation and deeper diagnostic analysis when finer-grained failure information becomes available.

The event logging process, illustrated in Figure 2, includes the following components:

- **Location_unit**: hierarchical mapping of plant areas.
- **Asset**: inventory of hydraulic equipment linked to locations.
- **Calendar_dim**: temporal dimension for aligning events with production schedules.
- **Event_log**: central table recording downtime events with asset IDs, timestamps, and related categories, problems, and causes.
- **Event_category**, **Problem_type**, **Root_cause**: classification tables supporting consistent fault taxonomy.

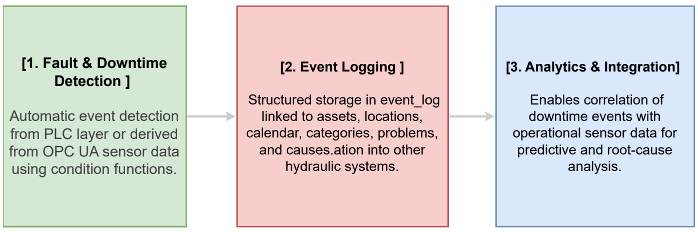


Fig. 2: Standard data collection behavior within the proposed framework. This diagram illustrates the normal acquisition and logging process for operational and fault events. For the detailed database organization and schema design of the logging system, refer to Appendix B.

Downtime events are categorized according to predefined operational taxonomies and associated with contextual sensor data (e.g., pressure, stroke length, temperature).

Table 6: Downtime Category Definitions

Category	Definition
Operational Standby	Equipment is available but idle due to external factors such as lack of upstream input or scheduling constraints.
Planned Maintenance	Scheduled inspections, calibration procedures, or preventive part replacement activities.
Unplanned Failure (Emergency)	Sudden production stop caused by unexpected system failures; in hydraulic systems this may include pressure loss, valve malfunction, or actuator blockage.
Resource Unavailability	Downtime caused by missing or insufficient resources, such as low oil levels, blocked supply lines, or unavailable utilities in hydraulic systems.

Note: The categorization follows common industrial downtime classification schemes and is adapted to hydraulic and fluid power systems where applicable.

Rather than constituting a physical failure ground truth, these records provide a consistent and reproducible operational reference layer derived from PLC logic that captures why and under which conditions a machine transitioned into a non-operational state.

This first-layer event classification enables a clear separation between operational standby, planned maintenance, unplanned equipment-related stoppages, and external resource or upstream process unavailability. By explicitly distinguishing machine-internal issues from line-level or organizational constraints, the database establishes a structured operational baseline for subsequent exploratory analysis and data-driven modeling.

The resulting structured event-log dataset supports operational root-cause investigation and the construction of predictive-maintenance targets without requiring explicit identification of detailed physical failure mechanisms. In this work, the event records are used as an operational reference layer for state comparison, RUL-target construction, and post-deployment monitoring of maintenance- and availability-related KPIs. The corresponding condition-level encoding, PLC event-code taxonomy, and representative event-log structure are reported in Appendix C..

Operational reference layer: PLC-Based Event Logging

In industrial practice, PLC- and supervisory-level event records constitute the primary operational trace for machine availability, stoppages, and alarm handling. In the present case study, these records are generated through deterministic PLC logic and cycle-based aggregation of key hydraulic variables. Here, a *hydraulic cycle* is defined as the smallest repeatable operational unit of the actuator under consideration. For linear actuators such as hydraulic cylinders, one cycle corresponds to a complete extension–retraction sequence. For rotary actuators, one cycle corresponds to one complete revolution or, more generally, to one complete repeatable motion between recurrent reference positions when the motion range is partial. The corresponding thresholds and event definitions were specified using hydraulic domain knowledge and adapted to the observed behavior of the investigated system.

This cycle-level representation is particularly relevant for PLC-based monitoring because embedded controllers typically do not retain long high-frequency histories for continuous comparison. Instead, they evaluate aggregated descriptors of the current cycle against predefined thresholds or recent prior cycles, providing a computationally efficient abstraction for detecting non-nominal behavior under practical memory and processing constraints.

Because the PLC logging layer is based on observable process signals and predefined operational rules, it does not constitute a physical ground truth of degradation. Instead, it provides a consistent and reproducible operational reference layer that captures when and under which observable conditions the system entered a non-nominal state. The resulting logs are therefore used as structured operational context for labeling, correlation, and model evaluation, while acknowledging that some degradation mechanisms may remain only partially observable due to sensing limitations.

4.2 ETL Architecture for Sensor Data Acquisition

The proposed ETL (Extract, Transform, Load) framework, whose architecture is depicted in Figure 3, enables continuous acquisition of time-series sensor data from OPC UA-enabled devices. Raw measurements are ingested in real time, transformed into a standardized format, and stored in normalized relational tables suitable for querying, visualization, and machine learning.

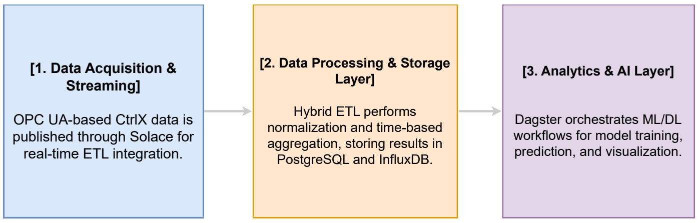


Fig. 3: Conceptual overview of the data streaming and ETL architecture underlying the proposed research framework. The diagram highlights the main stages from acquisition to storage and analytics. A detailed orchestration scheme is provided in Appendix B.

As described in Algorithms 1, 2, and 3, all components rely on open-source technologies, ensuring adaptability and eliminating licensing constraints. The messaging backbone is built upon Solace PubSub+, available in its Standard Edition at no cost [57].

4.2.1 Rationale for Hybrid Time-Series and Relational Storage

While time-series databases such as InfluxDB are well suited for high-frequency sensor ingestion and exploratory visualization, their variable-centric dashboards (e.g., Grafana panels per signal) provide limited support for holistic asset-level reasoning when the number of signals and monitored assets scales. In large-scale industrial deployments, this often results in fragmented insights that hinder rapid assessment and decision-making by end users.

To address this limitation, the proposed architecture deliberately combines a non-relational time-series store with a relational data model derived through scheduled pivoting operations, as formalized in Algorithms 1, 2 and 3. Raw measurements are preserved in InfluxDB to retain full temporal fidelity, while frequency-aligned relational tables are generated to enable efficient aggregation, reporting, and machine-learning workflows.

It is emphasized that the selected technologies represent one possible realization rather than the only viable implementation. Messaging topics, scheduling mechanisms, and orchestration layers may be instantiated through alternative solutions depending on deployment constraints. The architectural principle—decoupling raw data fidelity from analysis-ready representations—remains invariant.

This work emphasizes a transparent and interpretable data engineering methodology that balances computational efficiency with analytical value. By explicitly structuring data transformations and aggregation layers, the proposed approach avoids treating industrial analytics as a black box and enables domain experts to relate aggregated indicators to underlying physical phenomena.

Beyond its infrastructural role, the proposed ETL framework is designed to explicitly support degradation-aware learning. The multi-resolution aggregation strategy preserves fast transient phenomena at high sampling rates while providing statistically stable representations at coarser scales, which are required for reliable RUL estimation. This separation enables prognostic models to disentangle short-term dynamics from long-term wear evolution without loss of physical interpretability.

Algorithm 1 Edge-Oriented ETL Framework for PLC-Integrated Hydraulic Systems (Part 1: Initialization & Extraction)

Require: Real-time sensor data from `ctrlX CORE` (via OPC UA)
Ensure: Pivoted production tables in PostgreSQL and a cloud-ready data lake
Phase 1: Service Initialization
1: Start Docker containers:
- PostgreSQL with `pg_cron` extension
- Solace Broker (MQTT)
- InfluxDB time-series database
- Telegraf metrics collector
- Dagster Orchestrator
- ETL Microservices (Extract, Transform, Load)
- Grafana visualization platform

Phase 2: Data Extraction
2: Establish OPC UA connection
3: **if** simulation mode is active **then**
4: Connect `ETL-Extract` to the simulated CtrlX server
5: **else**
6: Connect to the physical CtrlX OPC UA server
7: **end if**
8: Publish raw sensor readings:
- Topic: `opcua/data`
- Format: `{sensor_id, value, timestamp}`

Algorithm 2 Edge-Oriented ETL Framework (Part 2: Transformation & Loading)

Phase 3: Data Transformation

1: Subscribe to the `opcua/data` topic
- Apply value scaling (e.g., 0-100% to engineering units)

2: Publish processed data
- Topic: `ctrlx/transformed`
- Add quality flags to the payload

Phase 4: Data Loading

3: Perform batch processing
- Buffer incoming records in 5-minute windows
- Bulk insert into the PostgreSQL long table
- Schema: `(sensor_id, value, timestamp, quality)`

Algorithm 3 Edge-Oriented ETL Framework (Part 3: Pivoting & Monitoring)

Phase 5: Data Pivoting

1: Schedule pivoting jobs
- `pg_cron` triggers at 100ms, 1s, and 1min intervals
- Generate frequency-synchronized tables

Phase 6: Orchestration & Monitoring

2: Provide system oversight
- Dagster monitors service health
- Automatic restart on failure detection

3: Collect metrics
- Use Telegraf → InfluxDB → Grafana pipeline
- Track latency, throughput, and error rates

4: Perform optional cloud integration
- Parallel write to a data lake
- Enable scalable analytics

Table 7 presents a representative data output with a 1-minute temporal resolution. In practice, the pipeline maintains three parallel output tables corresponding to aggregation frequencies of 100 ms, 1 s, and 1 min. These are refreshed every 5 minutes to consolidate recent activity into structures optimized for downstream processing.

To preserve diagnostic fidelity, raw acquisition is conducted at 100 ms intervals—essential for capturing fast transients such as internal leakage, stick-slip events, or abrupt pressure changes. These phenomena are often imperceptible at coarser resolutions but are critical indicators of early-stage degradation relevant for prognostic models such as RUL estimators.

The architecture is designed to be vendor-agnostic at the interface level, because it relies on OPC UA-based signal access and configurable signal mappings. However, transfer to other machines requires adaptation of the monitored variables, event rules, and operating-state definitions. Its configurable signal mapping and multi-resolution design support future extension to other applications, subject to system-specific validation, from high-speed hydraulic systems to slower electromechanical assets.

In summary, the ETL framework acts as a foundational data-engineering layer for prototype-level prognostic modeling by bridging low-level sensor acquisition with structured, analysis-ready data representations.

Table 7: Example output table with 1-minute frequency.

Timestamp	ENER	ACT_VE	EFF_SY	p_A
2025-05-27 11:37:00	225.88	0.60	21.99	1.87
2025-05-27 11:35:00	225.38	-2.45	27.04	1.98
2025-05-27 11:34:00	225.23	1.20	35.30	0.07
2025-05-27 11:33:00	225.43	-4.65	27.99	2.05
2025-05-27 11:32:00	225.28	-4.75	31.84	2.04
2025-05-27 11:31:00	225.38	-5.38	30.77	2.12

Note: Similar structures apply for 100 ms and 1-second intervals.

5 Methodological Contribution of the State-Transition RUL Formulation

This work proposes a hybrid anomaly-to-critical-state progression modeling framework for hydraulic condition monitoring under limited failure-label availability. The central methodological contribution is the definition of remaining useful life (RUL) as an operational state-transition horizon rather than as a terminal component-failure time. This formulation is particularly relevant for industrial hydraulic systems, where explicit run-to-failure experiments are often unavailable, but high-frequency measurements and PLC-based operational event logs provide valuable evidence about normal, warning, and critical operating conditions.

In this formulation, classification and RUL estimation serve different purposes. State classification provides a discrete assessment of the current condition, whereas state-transition RUL estimates the remaining time or cycles before a relevant operational boundary is crossed. The latter therefore provides a planning-oriented quantity that can support maintenance timing and prioritization when terminal failure labels are unavailable.

The proposed framework combines unsupervised state discovery with domain-informed interpretation. Since explicit degradation or failure labels are not available during normal operation, data-driven models such as K-Means and Hidden Markov Models (HMMs) are used to infer latent operating states directly from the measured hydraulic signals. These states are not treated as self-explanatory labels. Instead, they are interpreted a posteriori by correlating their temporal evolution with PLC event logs, expert knowledge, and known hydraulic behavior. In this way, abstract cluster centroids and state signatures are linked to physically meaningful phenomena such as pressure deviations, leakage-related behavior, flow loss, delayed actuator response, or critical operating thresholds.

Before estimating RUL, the framework performs a temporal correlation analysis between operational time, inferred state sequences, and PLC-supported condition levels. This step identifies whether the system evolves from nominal behavior toward warning or critical states, for example through transitions of the form normal $\rightarrow$ warning $\rightarrow$ critical. RUL is then estimated as the remaining time, or equivalently the remaining number of hydraulic cycles, before the system leaves the nominal operating region and enters an abnormal state supported by the PLC-based operational reference layer.

This differs from conventional RUL formulations that require rare terminal failure events or complete lifetime trajectories. The proposed approach focuses instead on precursor states that may support maintenance decision-making. If actual failure events are available, they can naturally be incorporated to refine the estimated RUL boundaries. However, the framework does not depend on explicit failure examples as a prerequisite. This enables earlier intervention once deviations from nominal operation become detectable, while still remaining compatible with future failure-enriched datasets.

In the present study, two operational prognostic targets are distinguished:

- **State-transition RUL:** remaining time or number of cycles before the system transitions from nominal operation to a PLC-supported abnormal state.
- **Critical RUL:** remaining time or number of cycles before the system reaches a high-risk or critical operational condition.

5.1 Positioning with Respect to Existing RUL Definitions

RUL is commonly defined as the remaining time until a degradation process reaches a predefined failure criterion. However, the meaning of this criterion depends on the available observability, the monitored component, and the intended maintenance decision. In hydraulic and electro-hydraulic systems, previous studies have used different forms of degradation evidence to support health-state assessment or RUL estimation.

Duensing et al. [32] showed that load-holding states in electro-hydrostatic actuators provide diagnostically relevant information about axial piston pump degradation. Wear-related effects were reflected in slower pressure build-up, increased rotational speed demand, internal leakage, pressure fluctuations, and efficiency losses. This demonstrates that operational states can encode degradation-relevant hydraulic information even without explicit run-to-failure labels. However, their work focuses on degradation identification and health-state interpretation rather than on a direct probabilistic RUL formulation.

Other hydraulic RUL approaches rely on explicit lifetime or event-time modeling. Li et al. [58] proposed a Bayesian joint model for critical hydraulic components, where condition monitoring, inspection, and event data are combined to estimate a conditional reliability function. In that formulation, RUL is derived from the reliability function of the monitored sample, conditioned on survival up to the current monitoring time and on the available inspection history. Such an approach is well suited when long-life tests, inspection records, and event-time data are available.

A related threshold-based interpretation is used in stochastic degradation models. Feng et al. [59] define RUL as the first hitting time at which a nonlinear degradation process reaches a predefined failure threshold. Their framework extends this concept by combining continuous degradation with discrete shock damage, thereby accounting for both gradual aging and sudden degradation contributions.

In contrast, the present work addresses a setting in which explicit component-level failure times are not available and PLC event logs are not treated as perfect physical ground truth. Instead, the logs are used as an operational reference layer that encodes expert-defined warning and critical conditions. Consequently, RUL is defined here as a *state-transition RUL*: the remaining time or number of hydraulic cycles before the system leaves the nominal state and enters a PLC-supported abnormal state. This shifts the prognostic target from terminal component failure to the earlier and more actionable transition from normal to warning or critical operation.

Table 8: Positioning of the proposed state-transition RUL definition with respect to related work.

Reference	**RUL / health criterion**	**Relation to this work**
Duensing et al. [32]	Load-holding-state degradation indicators for EHA axial piston pumps	Shows that operational states contain wear-related hydraulic information, but does not formulate RUL as a transition horizon.
Li et al. [58]	Conditional reliability based on monitoring, inspection, and event-time data	Provides a reliability-based RUL formulation when lifetime and event data are available.
Feng et al. [59]	First hitting time of a nonlinear degradation process reaching a failure threshold	Represents a threshold-based stochastic RUL definition with shock-induced damage.
This work	First transition from nominal to PLC-supported abnormal hydraulic operation	Defines an operational RUL for unlabeled industrial data, using PLC logs as an expert-informed reference layer rather than perfect failure labels.

5.2 State-Transition Remaining Useful Life

Let $x(t) \in \mathbb{R}^p$ denote the hydraulic feature vector observed at time t, and let $\hat{z}(t) \in \mathcal{S}$ denote the operating state estimated by the unsupervised state model, where $\mathcal{S}$ is the finite set of latent hydraulic states. The notation $\hat{z}(t)$ is used to emphasize that the state is inferred from measurements rather than directly observed. The true physical degradation state is not assumed to be available.

Since the raw state indices have no intrinsic operational meaning, a fixed mapping from latent states to PLC-referenced condition levels is defined as

$$g : \mathcal{S} \to \{1, 2, 3\}, \tag{13}$$

where level 1 denotes nominal operation, level 2 denotes warning or abnormal operation, and level 3 denotes critical operation. This mapping is based on centroid or state signatures, temporal behavior, hydraulic interpretation, and comparison with PLC-based operational reference logs. It is fixed before metric computation and is not optimized to maximize classification performance.

The nominal, abnormal, and critical state sets are defined as

$$\mathcal{S}_{\text{nom}} = \{s \in \mathcal{S} : g(s) = 1\}, \tag{14}$$

$$\mathcal{S}_{\text{abn}} = \{s \in \mathcal{S} : g(s) \geq 2\}, \tag{15}$$

and

$$\mathcal{S}_{\text{crit}} = \{s \in \mathcal{S} : g(s) = 3\}. \tag{16}$$

Thus, $\mathcal{S}_{\text{crit}} \subset \mathcal{S}_{\text{abn}}$, and $\{\mathcal{S}_{\text{nom}}, \mathcal{S}_{\text{abn}} \setminus \mathcal{S}_{\text{crit}}, \mathcal{S}_{\text{crit}}\}$ forms a partition of the inferred state space.

The monitoring history available at time t_k is denoted by

$$\mathcal{H}_k = \big\{x(\tau) \in \mathbb{R}^p, \hat{z}(\tau) \in \mathcal{S}, y_{\text{PLC}}(\tau) \in \{1, 2, 3\} : \, 0 \leq \tau \leq t_k\big\}, \tag{17}$$

where $y_{\text{PLC}}(\tau)$ is the PLC-referenced operational condition level. The PLC label is used as an operational reference layer, not as perfect physical ground truth.

The state-transition RUL is defined only for monitoring instants at which the system is still in nominal operation, i.e.,

$$\hat{z}(t_k) \in \mathcal{S}_{\text{nom}}. \tag{18}$$

Under this condition, the first transition time into abnormal operation is

$$T_{\text{abn}}(t_k) = \inf\big\{t > t_k : \hat{z}(t) \in \mathcal{S}_{\text{abn}}\big\}, \tag{19}$$

and the state-transition RUL is

$$RUL_{\text{state}}(t_k) = T_{\text{abn}}(t_k) - t_k. \tag{20}$$

Analogously, the first transition time into critical operation is

$$T_{\text{crit}}(t_k) = \inf\big\{t > t_k : \hat{z}(t) \in \mathcal{S}_{\text{crit}}\big\}, \tag{21}$$

and the corresponding critical RUL is

$$RUL_{\text{crit}}(t_k) = T_{\text{crit}}(t_k) - t_k. \tag{22}$$

If a sample is already abnormal, RUL_{state} is set to zero. If a sample is already critical, RUL_{crit} is set to zero. Sessions or time intervals without a future observed transition before the end of the acquisition window are treated as right-censored and are not used for deterministic regression target construction in the present evaluation.

Regression target construction.

For supervised RUL estimation, transition times are constructed retrospectively from the inferred HMM state sequence within each experimental session. For a session s with chronologically ordered samples $\{(x(t_i), t_i)\}_{i=1}^{N_s}$, let $T_{\text{abn}}^{(s)}$ and $T_{\text{crit}}^{(s)}$ denote the first observed transition times into $\mathcal{S}_{\text{abn}}$ and $\mathcal{S}_{\text{crit}}$, respectively. The regression targets are then defined as

$$y_1(t_i) = T^{(s)}_{\text{abn}} - t_i, \qquad t_i < T^{(s)}_{\text{abn}}, \tag{23}$$

and

$$y_c(t_i) = T^{(s)}_{\text{crit}} - t_i, \qquad t_i < T^{(s)}_{\text{crit}}. \tag{24}$$

Here, y_1 denotes the target for the first transition from nominal to abnormal operation, while y_c denotes the target for the transition to critical operation. Only positive target values are retained for regression training and evaluation.

A supervised regressor

$$\hat{f}_\theta : \mathbb{R}^p \to \mathbb{R}_+ \tag{25}$$

is then trained to approximate the operational RUL target from the current hydraulic feature vector:

$$\hat{f}_\theta(x(t_k)) \approx RUL_{\text{state}}(t_k). \tag{26}$$

For the first-transition target, the model parameters are obtained by minimizing

$$\hat{\theta} = \arg\min_\theta \sum_{(x_i, y_{1,i}) \in \mathcal{D}^{\text{tr}}} \left(y_{1,i} - \hat{f}_\theta(x_i) \right)^2 + \Omega(\theta), \tag{27}$$

where $\mathcal{D}^{\text{tr}}$ denotes the training partition and $\Omega(\theta)$ is the model-specific regularization term. The same construction is applied to y_c for critical RUL estimation.

For completeness, a probabilistic interpretation can be written as

$$R_{\text{state}}(\Delta \mid \mathcal{H}_k) = \Pr\left(T_{\text{abn}}(t_k) - t_k > \Delta \mid \hat{z}(t_k) \in \mathcal{S}_{\text{nom}}, \mathcal{H}_k\right), \qquad \Delta > 0. \tag{28}$$

The expected state-transition RUL is then

$$\mathbb{E}\left[RUL_{\text{state}}(t_k) \mid \mathcal{H}_k\right] = \int_0^\infty R_{\text{state}}(\Delta \mid \mathcal{H}_k)\, d\Delta. \tag{29}$$

In the present study, however, the experimental evaluation focuses on deterministic regression estimates of RUL_{state} and RUL_{crit}.

Cycle-based formulation.

Because the investigated hydraulic press operates through repeated extension–retraction cycles, the same transition horizon can also be expressed in cycle units. Let c_k denote the current hydraulic cycle and let $\hat{z}_c$ be the state assigned to cycle c. The first abnormal and critical cycles are

$$C_{\text{abn}}(c_k) = \inf\{c > c_k : \hat{z}_c \in \mathcal{S}_{\text{abn}}\}, \tag{30}$$

and

$$C_{\text{crit}}(c_k) = \inf\{c > c_k : \hat{z}_c \in \mathcal{S}_{\text{crit}}\}. \tag{31}$$

The corresponding cycle-based RUL targets are

$$RUL^{\text{cycle}}_{\text{state}}(c_k) = C_{\text{abn}}(c_k) - c_k, \tag{32}$$

and

$$RUL^{\text{cycle}}_{\text{crit}}(c_k) = C_{\text{crit}}(c_k) - c_k. \tag{33}$$

The relation between time-based and cycle-based RUL can be approximated by

$$RUL^{\text{cycle}}_{\text{state}}(c_k) \approx \frac{RUL_{\text{state}}(t_k)}{\bar{T}_{\text{cyc}}}, \tag{34}$$

where $\bar{T}_{\text{cyc}}$ is the mean cycle duration estimated from previous completed cycles. This approximation is valid when cycle duration is approximately stationary. Under degradation-induced slowdown, cycle duration may increase; therefore, time-based RUL is retained as the primary regression target, while cycle-based RUL provides an operationally interpretable equivalent for maintenance planning.

6 Experimental Data and Machine Learning Protocol

ML and DL are commonly used tools for supporting data-driven condition monitoring and prognostics in industrial systems. By integrating real-time and historical sensor data, these approaches enable rapid visualization of system states and facilitate data-driven intervention strategies. However, their value must be rigorously measured—not only in terms of technical performance (e.g., accuracy, robustness) but also through broader economic impact indicators such as cost savings, operational efficiency gains, and enhanced system reliability.

For practical use of ML/DL models in industrial prognostics, three aspects are particularly relevant:

1. **Model Performance:** Achieve statistically robust accuracy and stability under varying operational conditions.
2. **Domain Alignment:** Ensure model outputs are consistent with expert knowledge and physically meaningful system behavior.
3. **Stakeholder Adoption:** Support acceptance from domain experts and decision-makers by providing interpretable and operationally relevant results.

6.1 Data Set Description

The experimental data were acquired from the PLC-controlled electro-hydraulic press testbed described in Section 3. The system was operated under repeated extension–retraction cycles using identical setpoints across all experimental sessions. Consequently, the repeated acquisitions were not intended to represent different operating regimes, but to quantify repeatability under nominally equivalent hydraulic operating conditions.

In total, 15 hydraulic sessions were recorded at a sampling resolution of 100 ms. The original raw dataset contains 620,648 samples and 20 measured variables. After signal cleaning, feature construction, and session-level enrichment, the analysis-ready dataset contains 618,672 samples and 28 variables. In parallel, PLC-oriented operational event logs were generated from threshold-based condition rules and manual contextual annotations, resulting in 1,439 final event records with 27 fields across all sessions. A summary of the dataset is provided in Table 9.

The experiment represents a controlled commissioning and prototype-level evaluation scenario using real industrial hydraulic hardware. Since long-term degradation mechanisms such as pump wear, internal leakage, and viscosity-related performance losses normally evolve over extended operating periods, thermal effects and controlled internal leakage were introduced to make degradation related performance changes observable within a feasible laboratory acquisition window.

Table 9: Summary of the experimental dataset used in this study.

Property	**Description**
Hydraulic system	PLC-controlled hydraulic press testbed
Operating profile	Repeated extension–retraction cycles
Number of sessions	15 hydraulic sessions
Operating setpoints	Constant across all sessions
Sampling resolution	100 ms
Raw sensor data	620,648 samples × 20 variables
Analysis-ready data	618,672 samples × 28 variables
PLC event logs	1,439 records × 27 fields
Degradation protocol	Accelerated leakage and thermal degradation
Training and evaluation basis	All 15 sessions
Validation reference	PLC-based operational event logs

Table 9 summarizes the complete data basis used in this study. The reduction from the raw to the analysis-ready dataset is caused by signal cleaning and invalid-sample removal, whereas the increase in the number of variables reflects the construction of derived hydraulic, cycle-level, and session-level features.

6.1.1 Repeated-Session Acquisition and Repeatability

Figure 4 compares the actuator velocity across the 15 repeated hydraulic sessions. The similar dynamic behavior confirms that the sessions are operationally comparable under the selected setpoints. In this study, all 15 sessions are retained for model training and evaluation in order to increase the amount of available degradation-relevant data and to expose the models to minor session-to-session variability. A representative session is used only for detailed visualization of state transitions and PLC-event distributions, not as the sole basis for model training.

To quantify repeatability, inter-session variability was computed using session-level mean and median deviations. As shown in Table 10, the average deviations remain below 1%, while the maximum deviations remain below 2.5%. This confirms that the sessions are not numerically identical, but are operationally equivalent under the investigated setpoints. The observed differences are therefore interpreted as minor sensor- and process-level variability rather than changes in operating regime.

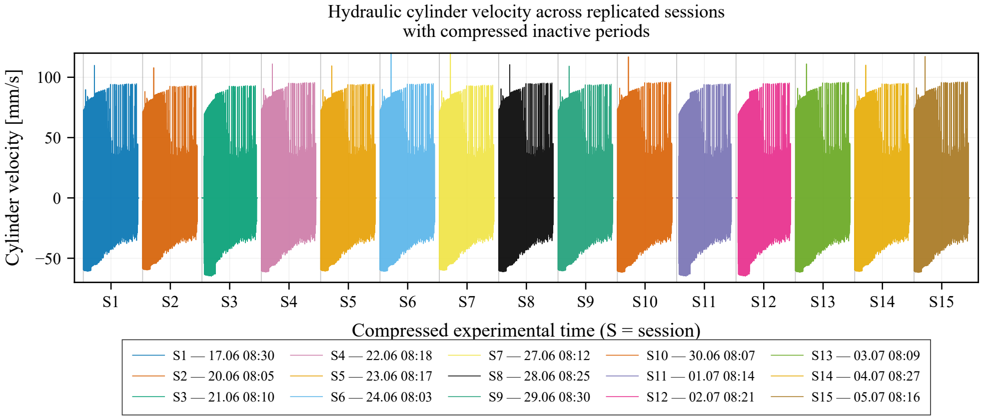


Fig. 4: Compressed visualization of actuator velocity across the 15 repeated hydraulic sessions. All sessions were acquired under identical operating setpoints. The limited inter-session variability supports the interpretation that the sessions are operationally comparable, while still retaining minor sensor- and process-level variation relevant for model evaluation.

Table 10: Inter-session variability under repeated operating conditions.

Metric	Mean	Median	Std.	Min.	Max.
Mean variation of session means (%)	0.60	0.60	0.23	0.13	0.94
Median variation of session means (%)	0.47	0.43	0.26	0.12	0.92
Mean variation of session medians (%)	0.64	0.61	0.25	0.12	1.00
Median variation of session medians (%)	0.52	0.48	0.31	0.12	1.18
Maximum mean-based deviation (%)	1.43	1.40	0.50	0.28	2.35
Maximum median-based deviation (%)	1.47	1.45	0.46	0.30	2.40

6.1.2 PLC-Based Operational Reference Logs

The PLC event logs are used as an operational reference layer rather than as perfect physical ground truth. They reflect how the machine condition was observed from the industrial control perspective, combining PLC-level threshold logic with manual contextual annotations. This distinction is important because the machine states are inferred from sensor-derived features, while the PLC logs provide an independent operational reference for evaluating whether the inferred states are consistent with recorded machine behavior.

Table 11 reports the distribution of operational reference events for a representative degradation session. This session was selected for detailed visualization because it contains the complete progression from normal operation toward warning and critical hydraulic behavior, together with corresponding PLC-based operational event coverage. Manual contextual reports are kept separate because they provide operational context rather than automatic condition-level labels.

Table 11: Distribution of PLC operational reference events in the representative session.

Condition class	Condition level	Intervals	Duration (min)
Normal operation	1	15	68.23
Warning operation	2	32	25.04
Critical operation	3	28	137.62
Manual contextual reports	–	6	89.00

Note: Automatic events are grouped by condition level, whereas manual contextual reports are listed separately. Overlapping records from different source variables are retained because they represent distinct PLC-log entries.

The complete event-code taxonomy used to generate the operational reference labels is reported in Appendix C. This keeps the main text focused on the dataset structure while preserving reproducibility of the state-comparison and RUL-target construction process.

6.1.3 Hydraulic-Cycle Representation

Because the investigated system operates through repeated extension–retraction strokes, the data were also analyzed at hydraulic-cycle level. Each hydraulic cycle represents one complete extension–retraction sequence of the actuator. This cycle-level representation is relevant for both PLC-based monitoring and RUL estimation, because degradation in press-like hydraulic systems often becomes visible as changes in cycle duration, velocity extrema, and cycle-to-cycle repeatability.

Figure 5 shows the temporal evolution of the cycle duration, while Table 12 summarizes its descriptive statistics across all analyzed cycles. The cycle duration has a median of 30.30 s and an interquartile range from 22.70 s to 36.40 s, indicating that most cycles remain within a consistent operating range, while longer durations reflect degraded or slower operating phases.

Table 12: Descriptive statistics of the hydraulic cycle duration across all analyzed samples.

Statistic	Hydraulic cycle duration (s)
Valid samples	610,777
Mean	32.62
Std. deviation	13.64
Minimum	3.47
25th percentile	22.74
Median	30.45
75th percentile	37.01
Maximum	83.90

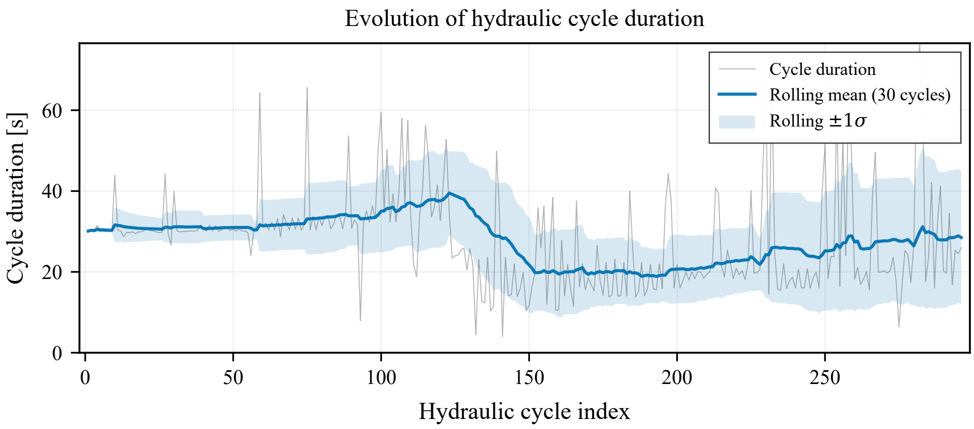


Fig. 5: Evolution of the hydraulic cycle duration across the analyzed extension–retraction cycles. Each point represents one complete hydraulic cycle. The rolling mean and rolling standard deviation summarize local temporal variability and reveal changes in cycle repeatability over time.

6.1.4 Data Cleaning and Analysis-Ready Dataset

To prevent bias in the statistical characterization caused by sensor dropouts, interpolation artifacts, or physically implausible actuator behavior, the raw measurements were processed through a structured data cleaning pipeline, summarized in Algorithm 4. The objective of this procedure is not aggressive denoising, but the preservation of physically meaningful hydraulic dynamics while excluding invalid samples. This reduces the risk of introducing artificial degradation trends caused by interpolation artifacts or physically implausible measurements.

Algorithm 4 Data Cleaning Process (Class `DataCleaner`)

1: Load configuration parameters, including NaN thresholds and smoothing windows
2: Interpolate missing values for signals below the predefined NaN-ratio threshold
3: Clean velocity signal `CtrlX_2_ACT_VE`:
    interpolate missing values, smooth signal, and detect abnormal peaks
4: Clean position signal `CtrlX_2_ACT_PO`:
    remove outliers and apply smoothing
5: Identify and remove invalid samples based on pressure and actuator-motion constraints
6: Construct derived hydraulic, cycle-level, and session-level features
7: Store the resulting analysis-ready dataset for state modeling and RUL estimation

Figure 6 illustrates the empirical distributions of the raw variables used in this study. These distributions provide an overview of the signal ranges before model-specific feature selection. All subsequent classification and RUL regression analyses operate on the cleaned and analysis-ready dataset.

For both state classification and RUL regression, the cylinder velocity, pressure ratios, oil flow rate, cylinder position, and cycle-level descriptors are emphasized because they are directly related to the hydraulic degradation mechanisms investigated

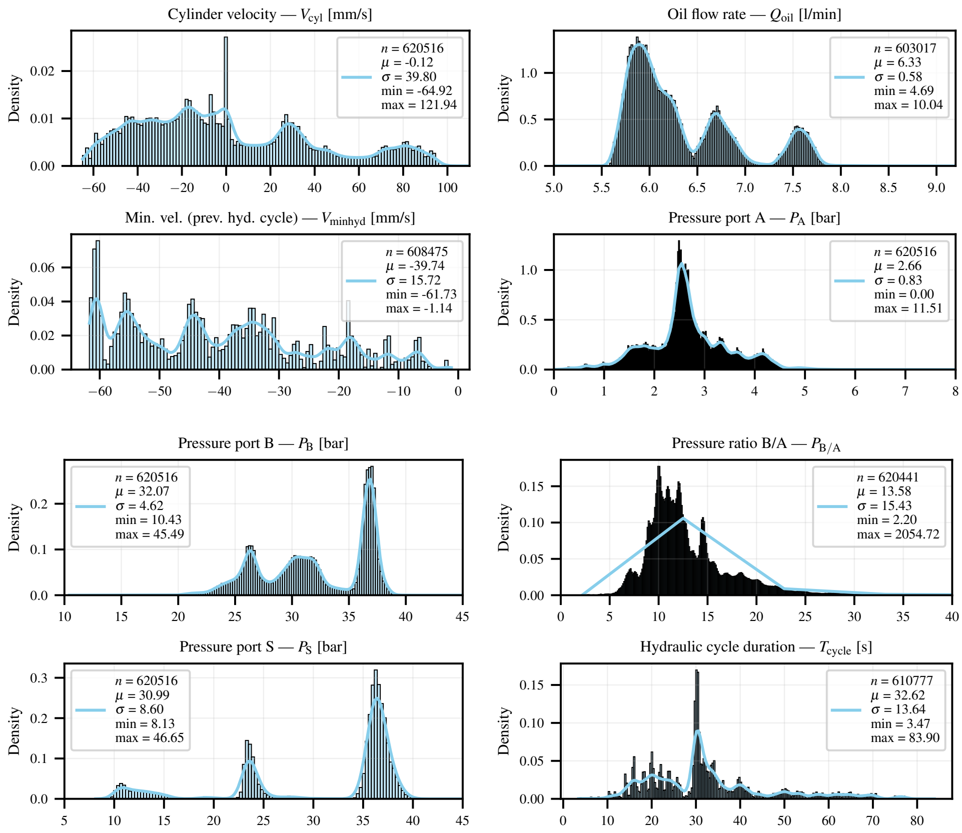


Fig. 6: Empirical distributions of raw variables, including histograms, KDE, and statistical summaries across four variable sets.

in this study. In particular, retraction velocity and cycle duration capture slowdown effects caused by leakage and thermal performance loss, while pressure ratios provide information about chamber imbalance and pressure-loss behavior.

Overall, the dataset combines high-frequency sensor measurements, repeated hydraulic sessions, cycle-level descriptors, and PLC-based operational event logs. This structure enables the subsequent evaluation of unsupervised state identification and state-transition RUL estimation from both a signal-driven and an operational-reference perspective.

6.2 Machine State Classification

For condition monitoring and anomaly detection, the unsupervised clustering algorithm K-Means was employed to identify abnormal patterns. According to Chandola et al. [60], clustering-based methods are widely recognized as effective techniques for anomaly detection. Building on this foundation, Hidden Markov Models (HMM) were additionally applied to detect operational states. These models were supported by preprocessing techniques that calculated signal variation, oscillation, and envelope characteristics over time to distinguish critical operating states.

6.2.1 State-Model Feature Engineering

Data preprocessing followed the methodology outlined in Algorithm 5. One of the key challenges, as observed in the ETL output table (Section 4.2), is the high-frequency variability inherent to hydraulic systems. Due to their fast dynamic response, variables such as pressure, position, and flow rate can exhibit significant changes within milliseconds — for instance, cylinder velocity, expressed in mm/s, fluctuates rapidly.

In the investigated setup, degradation-related performance changes are reflected primarily through changes in the consistency and repeatability of motion rather than absolute signal levels. In short-term observation windows, these effects are more accurately described as performance degradation, reflecting a loss of dynamic efficiency rather than irreversible mechanical wear.

Internal leakage and viscosity reduction alter the force balance within the cylinder, leading to increased cycle-to-cycle variability in both position and velocity signals. As degradation progresses, these effects appear as (i) gradual widening of the signal envelopes, (ii) increased oscillation variance within each stroke, and (iii) a progressive increase in the cycle period due to reduced effective actuation efficiency.

Envelope-based representations therefore provide a compact means of capturing both transient performance degradation and longer-term trends, while oscillation and cycle-based features quantify short-term dynamic irregularities that may precede or accompany hydraulic wear.

Algorithm 5 Feature Engineering Process (Class `DataProcessor`)

1: Compute oscillation metrics from cleaned position signal
2: Detect position peaks and extract cycle-based amplitude and period
3: Compute rolling statistics (mean, std) of amplitude and period features
4: Estimate velocity and position envelopes and apply smoothing
5: Suppress amplitude and period features outside inverse velocity peak regions

To assess the effect of temporal resolution on the visibility of degradation-relevant hydraulic behavior, the proposed ETL framework preserved parallel acquisition and aggregation layers at 100 ms, 1 s, and 1 min. Comparative inspection of these representations showed that, for the present hydraulic setup, the 100 ms layer retained the short-term variations required to distinguish localized velocity anomalies, leakage-related irregularities, and cycle-to-cycle dynamic changes, whereas the coarser resolutions were more suitable for reporting and long-horizon aggregation. Accordingly, 100 ms was adopted as the working resolution for feature extraction and state-based modeling.

In particular, the velocity signal exhibits localized anomalies, visible as small amplitude increases, that become distinguishable only in the higher-resolution representation (Figure 8). These short-term variations provide relevant information for the RUL estimators evaluated in this work. Figures 7 present the derived features analyzed within a moving-window framework. Each cycle corresponds to a full stroke of the hydraulic cylinder. Over time, both the position signal and its derivative (velocity) exhibit variations in amplitude and period that are consistent with evolving performance degradation in the investigated setup. In the present setup, these variations are primarily associated with internal leakage and dynamic limitations that reduce the system's ability to maintain repeatable motion profiles.

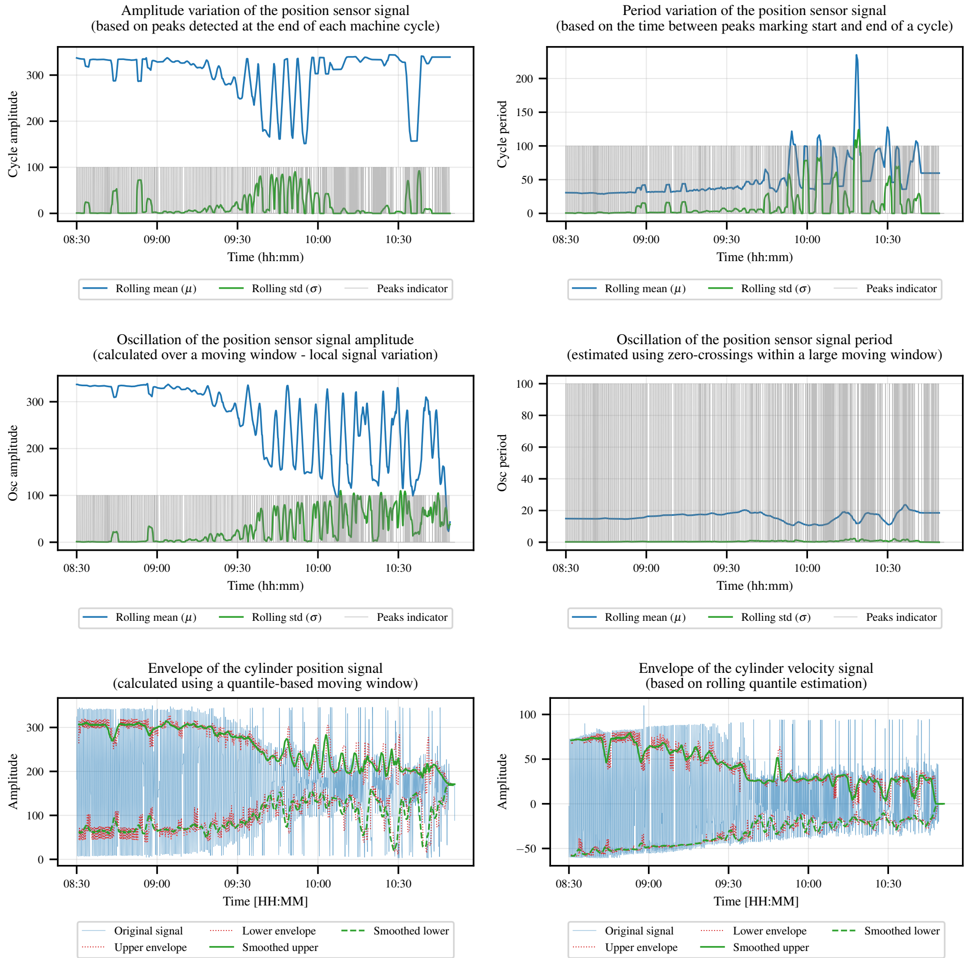


Fig. 7: Extracted features derived from the hydraulic cylinder position and velocity signals. Rolling metrics characterize cycle- and oscillation-level dynamics, while envelope-based features capture amplitude variability over time.

Figure 8 shows an excerpt of three extracted variables over a minute-scale interval. The signals exhibit rapid oscillatory behavior, which is characteristic of hydraulic systems operating under closed-loop control and high-pressure conditions.

The signal denoted as L_{alarm} corresponds to a controlled leakage alarm introduced during the preprocessing stage. This leakage is manually simulated by opening valve Y3 in the hydraulic circuit (see Fig. 1), allowing the evaluation of detection sensitivity under well-defined fault conditions.

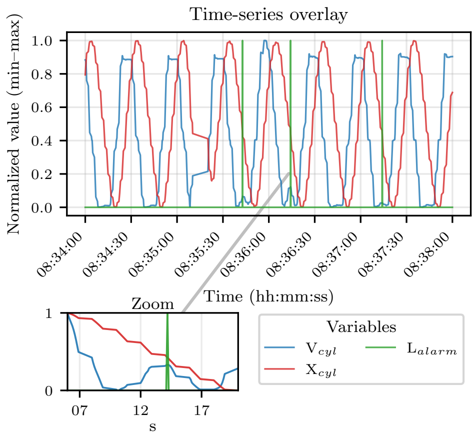


Fig. 8: Excerpt of selected hydraulic time-series signals.

In Figures 7, the envelope of the variable was determined using upper quartiles. This approach was adopted to evaluate variability, as it characterizes the amplitude fluctuations of the signal over time.

After analyzing the variables, those listed in Table 13 are used for the classification process.

Table 13: Input features used in clustering and state modeling.

Variable Name	Description
`osc_amplitude_rolling_mean`	Rolling mean of the oscillation amplitude.
`vel_cylinder_envelope_upper_smooth`	Smoothed upper envelope of the cylinder velocity signal.
`vel_cylinder_envelope_lower_smooth`	Smoothed lower envelope of the cylinder velocity signal.
`osc_period_rolling_mean`	Rolling mean of the oscillation period based on zero-crossing detection.
`cycle_period_rolling_mean`	Estimated period between consecutive peaks representing full cylinder cycles.
`pos_clean_envelope_lower_smooth`	Smoothed lower envelope of the cleaned cylinder position signal.

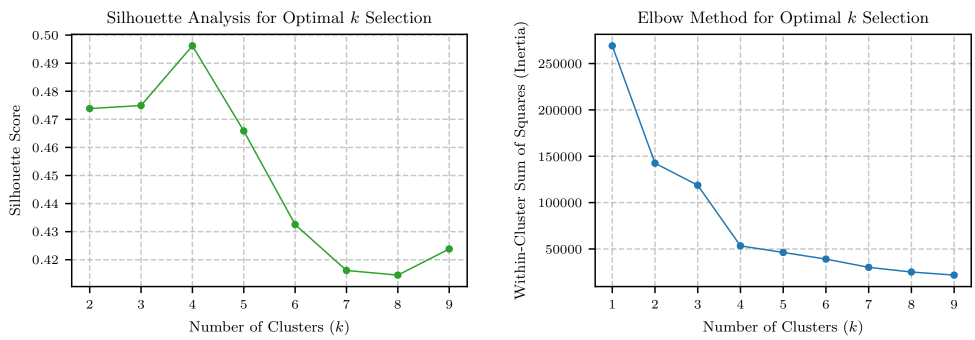
Note: All features are computed from cleaned signals and normalized prior to clustering and state modeling.

To assess unsupervised state segmentation from complementary perspectives, two conceptually distinct models are employed. K-Means clustering is used as a baseline to evaluate the *geometric separability* of operational states in the feature space, independent of temporal ordering. In contrast, the Hidden Markov Model (HMM) explicitly models *temporal coherence* and state persistence, which is critical for stable RUL estimation. Together, both methods provide a benchmark comparison between instantaneous geometric clustering and sequence-aware probabilistic state modeling.

6.2.2 Model 1: K-Means for State Clustering

K-Means was used to assess whether the engineered hydraulic descriptors form geometrically separable operating regimes in the normalized feature space. Unlike the HMM, which imposes temporal persistence through transition probabilities, K-Means provides an instantaneous state partition based only on feature similarity. This makes it suitable as a first diagnostic layer for evaluating whether the selected cycle-synchronous features contain sufficient information to separate normal, abnormal, critical, and standstill-like behavior.

The number of clusters was not selected only from the elbow curve. Instead, a multi-criteria model-selection analysis was conducted for $k = 2, \ldots, 9$, combining compactness, separation, and cluster validity metrics. Specifically, we evaluated the within-cluster sum of squares (inertia), the silhouette score, the Calinski–Harabasz index, and the Davies–Bouldin index. The elbow curve indicates a strong reduction in inertia up to approximately $k = 4$, after which the improvement becomes progressively smaller. In parallel, the silhouette score reaches its maximum at $k = 4$ (0.496), indicating the best trade-off between intra-cluster compactness and inter-cluster separation among the evaluated alternatives. Although the Davies–Bouldin index is lowest for $k = 2$, this solution merges physically distinct operating conditions and is therefore insufficient for the proposed state-transition RUL formulation. Conversely, values $k > 4$ provide only marginal additional compactness while reducing silhouette quality and fragmenting operationally interpretable regimes.



(a) Silhouette analysis.　　(b) Elbow method.

Fig. 9: Multi-criteria selection of the number of K-Means clusters. The elbow curve shows diminishing inertia reduction after $k = 4$, while the silhouette score reaches its maximum at $k = 4$, supporting the selection of four latent operating states.

Table 14: Cluster-validity metrics used for selecting the number of K-Means states.

k	**Inertia**	**Silhouette**	**Calinski–Harabasz**	**Davies–Bouldin**
1	249036.00	–	–	–
2	126637.90	0.474	40114.46	0.859
3	94174.77	0.475	34123.83	1.159
4	73790.71	**0.496**	32854.32	0.978
5	61611.91	0.466	31561.68	1.000
6	50692.04	0.433	32475.72	0.937
7	47694.31	0.416	29198.20	0.976
8	38324.52	0.415	32594.39	0.903
9	32671.14	0.424	34351.80	0.879

Note: All metrics were computed after standard scaling of the six state-modeling features. Higher silhouette and Calinski–Harabasz values indicate better separation, whereas lower inertia and Davies–Bouldin values indicate more compact clusters.

The selected value $k = 4$ was therefore retained because it satisfies three conditions: (i) it lies at the main elbow of the inertia curve, (ii) it maximizes the silhouette score among all tested configurations, and (iii) it yields the smallest number of states that can be mapped to hydraulically meaningful operating regimes required by the proposed RUL formulation. This choice also avoids over-partitioning the data into clusters that are statistically distinct but operationally redundant.

The clustering algorithm is summarized in Algorithm 6, and the corresponding configuration is reported in Table 15.

Algorithm 6 K-Means Clustering and Cluster-Number Validation

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{T\times 6}$, candidate set $\mathcal{K} = \{2, \dots, 9\}$
Ensure: Cluster assignment vector $\mathbf{z} \in \{0, \dots, k-1\}^T$
1: Normalize $\mathbf{X}$ using standard scaling
2: **for** $k \in \mathcal{K}$ **do**
3: Fit K-Means with k clusters using `k-means++` initialization
4: Compute inertia, silhouette score, Calinski–Harabasz index, and Davies–Bouldin index
5: **end for**
6: Select $k = 4$ based on the joint elbow–silhouette criterion and operational interpretability
7: Solve

$$\min_{\{\mathcal{C}_j\}_{j=1}^k} \sum_{j=1}^{k} \sum_{\mathbf{x}_t \in \mathcal{C}_j} \|\mathbf{x}_t - \boldsymbol{\mu}_j\|_2^2$$

8: Assign each $\mathbf{x}_t$ to its nearest centroid:

$$z_t = \arg \min_{j\in\{1,\dots,k\}} \|\mathbf{x}_t - \boldsymbol{\mu}_j\|_2^2$$

9: Store the trained scaler, centroids, and cluster-to-state mapping
10: **return z**

The selected features are low-dimensional, cycle-synchronous descriptors that emphasize dynamic variability rather than absolute signal levels. They capture oscillation amplitude, velocity-envelope widening, cycle-period variation, and position envelope changes, which are expected indicators of hydraulic performance degradation under leakage and thermal slowdown conditions. Consequently, the resulting clusters are interpreted not as supervised fault labels, but as latent operational states whose semantic meaning is assigned post hoc through their centroid signatures, temporal ordering, agreement with HMM state persistence, and comparison with the PLC-based operational reference layer.

Table 15: K-Means clustering configuration.

Parameter	**Value / Description**
Candidate cluster range	$k = 2, \ldots, 9$, systematically evaluated using inertia, silhouette, Calinski–Harabasz, and Davies–Bouldin scores.
Selected number of clusters	$k = 4$, selected from the joint elbow–silhouette criterion and retained as the minimum state resolution required for the state-transition RUL formulation.
Initialization method	`k-means++` for stable centroid initialization.
Distance metric	Euclidean distance in standardized feature space.
Optimization algorithm	`Elkan` algorithm, efficient for dense and low-dimensional data.
Random seed	Fixed seed (2025) for reproducible final model training.
Model persistence	The fitted scaler, centroids, and cluster-to-state mapping are stored and reused during inference.

Note: The cluster labels are not treated as physical ground truth. They are interpreted a posteriori by combining centroid analysis, temporal consistency with HMM states, and comparison with PLC-referenced operational events.

This stability analysis supports the numerical reproducibility of the clustering result, but it does not by itself constitute physical validation of the latent states; the hydraulic interpretation is therefore based on centroid signatures, temporal behavior, HMM consistency, and comparison with PLC-based operational reference logs.

K-Means seed stability.

The selected K-Means configuration was repeated for $k = 4$ using 50 random seeds. The resulting partitions were nearly identical across runs, with a mean ARI of 0.9999 and a mean NMI of 0.9996. The silhouette score also remained practically unchanged (0.4962 ± 0.00001). These results indicate that the four-state partition is numerically stable with respect to random initialization. Physical interpretation of the states is therefore based not on seed stability alone, but on centroid signatures, temporal behavior, HMM consistency, and comparison with PLC-based operational logs.

Table 16: K-Means stability across 50 random seeds for $k = 4$.

Metric	Mean	Std.	Range
ARI	0.999907	0.000075	0.999678–1.000000
NMI	0.999588	0.000325	0.998571–1.000000
Silhouette score	0.496195	0.000010	–
Inertia	73790.709936	0.010691	–
Davies–Bouldin index	0.977788	0.000013	–

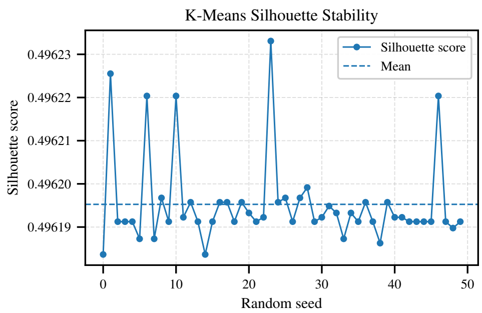


Fig. 10: Silhouette-score stability of the selected K-Means configuration across 50 random seeds for $k = 4$.

6.2.3 Model 2: HMM for State Classification

For the definition of states in the HMM, four states were employed. Similar to the cluster configuration, This setup provided a sequence-aware comparison to the differentiation obtained with K-Means and evaluated machine states and operating conditions, with the algorithm and model parameters summarized in Algorithm 7 and Table 17.

Algorithm 7 Hidden Markov Model for State Estimation

Require: Normalized feature matrix $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$, $\mathbf{x}_t \in \mathbb{R}^6$
Ensure: Most probable hidden state sequence $\mathbf{s} = (s_1, \ldots, s_T)$
1: Initialize a Gaussian HMM with $n = 4$ hidden states
2: Define state transition matrix $\mathbf{A}$ and initial distribution $\boldsymbol{\pi}$
3: Specify Gaussian emission densities $p(\mathbf{x}_t \mid s_t = i) = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ with diagonal covariance $\boldsymbol{\Sigma}_i$
4: Estimate model parameters $\{\mathbf{A}, \boldsymbol{\pi}, \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i\}$ by maximizing the log-likelihood

$$\log p(\mathbf{X} \mid \lambda) = \sum_{t=1}^{T} \log \left(\sum_{s_t} p(\mathbf{x}_t \mid s_t)\, p(s_t \mid s_{t-1}) \right)$$

using the Baum–Welch algorithm (maximum 1000 iterations)
5: Compute the most probable hidden state sequence $\mathbf{s}^\star = \arg\max_{\mathbf{s}} p(\mathbf{s} \mid \mathbf{X})$ using the Viterbi algorithm
6: Store trained HMM model
7: **return** $\mathbf{s}^\star$

Table 17: Hidden Markov Model configuration.

Parameter	Value / Description
Number of hidden states (n)	$n = 4$, representing discrete operational regimes inferred from the feature space.
Observation model	Multivariate Gaussian emission densities $p(\mathbf{x}_t \mid s_t = i) = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$.
Covariance structure	Diagonal covariance matrices $\boldsymbol{\Sigma}_i$, assuming conditional independence between features.
Training algorithm	Baum–Welch algorithm for maximum-likelihood parameter estimation.
Maximum iterations	Limited to 1000 iterations to ensure convergence of the log-likelihood.
Decoding algorithm	Viterbi algorithm for most probable hidden state sequence inference.
Random seed	Fixed seed (2025) to ensure reproducibility of training and decoding.

Note: The HMM operates on normalized features and estimates discrete system states by probabilistic sequence modeling.

6.3 Remaining Useful Life Estimation

To reduce circular interpretation, the PLC labels were not used to train the state models. In addition, the mapping from latent states to PLC levels was fixed before metric computation and was not optimized to maximize classification performance.

Table 18: Input features used for XGBoost-based RUL prediction.

Symbol	Unit	Description
X_{cyl}	mm	Cleaned cylinder position after interpolation and outlier removal.
$P_{\mathrm{A/S}}$	–	Pressure ratio between port A and supply pressure, indicative of load, pressure-loss, and leakage-related effects.
$P_{\mathrm{B/A}}$	–	Pressure ratio between port B and port A, sensitive to chamber imbalance and internal leakage.
V_{cyl}	mm/s	Instantaneous cylinder velocity derived from the position signal.
Q_{oil}	l/min	Oil flow rate, reflecting hydraulic supply behavior and leakage-related flow changes.
$V_{\mathrm{prev}}^{\mathrm{min}}$	mm/s	Minimum cylinder velocity recorded during the previous completed hydraulic cycle. This lagged PLC-derived descriptor captures short-term loss of dynamic performance without using future information.
T_{cycle}	s	Duration of the previous completed hydraulic cycle, defined as the elapsed time required to complete one full cylinder extension–retraction cycle. This feature captures cycle-level slowdown while preserving causality.

RUL estimation is formulated as a supervised regression problem built on the previously identified operating states. Instead of only assigning the current sample to a normal, abnormal, or critical condition, the model estimates the remaining time until the next relevant state transition. This transforms state identification into a prognostic quantity that is more useful for maintenance planning, because it provides an estimated intervention horizon rather than only an instantaneous condition label.

The input variables used for RUL prediction, listed in Table 18, are available from the sensor infrastructure and include pressure, flow, velocity, position, and cycle-level descriptors. The proposed RUL formulation is illustrated in Figure 11, where RUL and critical RUL are expressed as the remaining time to abnormal and critical operating states, respectively.

Baseline selection rationale.

The selected models were intended to compare complementary assumptions under the same temporal evaluation protocol. Linear and ridge regression serve as simple statistical trend baselines; XGBoost evaluates nonlinear interactions among interpretable hydraulic and cycle-level features; and LSTM-Attention evaluates whether temporal context across recent samples improves state-transition RUL prediction. Continuous health-indicator or degradation-path baselines were not used as primary models because the prototype does not provide complete run-to-failure trajectories or a monotonic component-level health indicator.

Visualization of the State-Transition RUL Concept

The proposed state-transition RUL is formally defined in Section 5.2. Figure 11 provides a graphical illustration of this formulation, where RUL and critical RUL are expressed as the remaining time to abnormal and critical operating states, respectively.

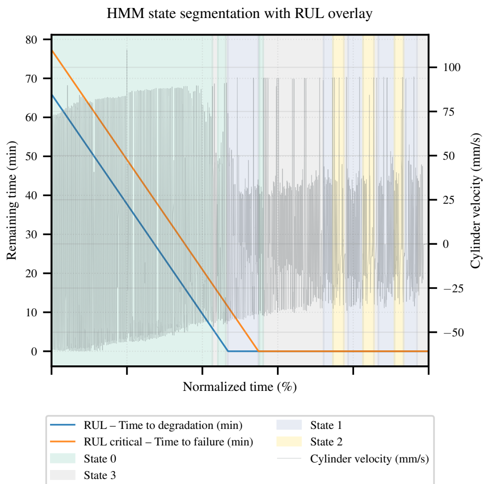


Fig. 11: RUL estimation based on state transitions.

6.3.1 Model 3: XGBoost Regressor for RUL Prediction

XGBoost was employed as a tabular regression baseline for state-based RUL estimation. In contrast to the sequence-based LSTM model, XGBoost operates directly on engineered hydraulic features and therefore provides a transparent benchmark for evaluating the predictive information contained in pressure, flow, velocity, position, and cycle-duration descriptors. Separate models were trained for the two prognostic targets: RUL and $RUL\_critical$. The configuration used for both targets is summarized in Table 19.

Table 19: XGBoost configuration for RUL prediction.

Parameter	**Value**
Model type	`XGBRegressor`
Objective function	`reg:squarederror`
Number of estimators	400 boosting rounds
Learning rate	0.02
Maximum tree depth	3
Minimum child weight	10
Subsample ratio	0.70
Column subsample by tree	0.50
Column subsample by node	0.50
L1 regularization	0.3
L2 regularization	7.0
Early stopping rounds	25, based on validation MAE
Evaluation metric	MAE
Parallel jobs	All available cores ($n_{\text{jobs}} = -1$)
Random seed	42

XGBoost: Target-aware temporal split.

To reduce information leakage, the XGBoost models were evaluated using a target-aware temporal split. For each RUL target and session, only samples within the valid prognostic region, i.e., $y(t) > 0$, were retained. The data were then sorted chronologically and divided into repeated mission blocks, where 70%, 10%, and 20% of each block were assigned to training, validation, and testing, respectively. Figure 12 illustrates this split strategy for one representative session.

Because all 15 sessions were acquired under nearly identical setpoints, this protocol evaluates whether the model can predict later degradation behavior from earlier observations within comparable missions, rather than only memorizing session-level repetition. Timestamps and session identifiers were used only for traceability and visualization, not as model inputs.

This split does not evaluate transfer to unseen machines or substantially different operating regimes. It evaluates temporally later degradation segments within repeated and comparable missions of the same prototype.

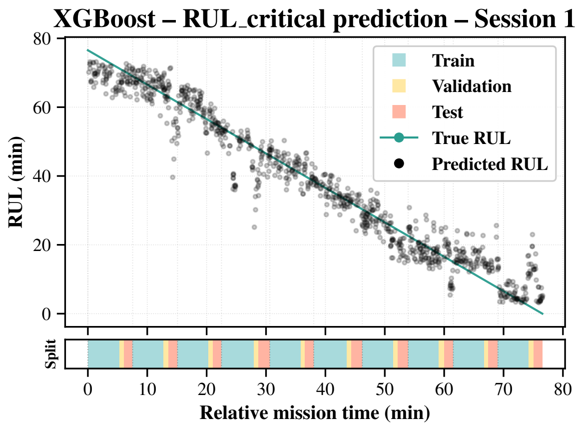


Fig. 12: Example of the target-aware temporal split for one experimental session. The upper panel compares true and predicted critical RUL, while the lower barcode shows the repeated train, validation, and test partitions across chronological mission blocks.

Algorithm 8 Target-aware XGBoost protocol for state-transition RUL prediction

---

Require: Cleaned dataset $\mathcal{D} = \{(\mathbf{x}_i, y_{1,i}, y_{2,i}, t_i, s_i)\}_{i=1}^{N}$, with targets $y_1 = \mathrm{RUL}$ and $y_2 = \mathrm{RUL}_c$
Ensure: Trained regressors $\hat{f}_1$, $\hat{f}_2$, test metrics, and post-hoc explanations

```
Define the hydraulic feature vector x ∈ ℝ^p from pressure, flow, velocity, position, and cycle-level descriptors.
Remove samples with missing values in x, y_1, y_2, t, or s.
Retain t and s only for temporal splitting, traceability, and visualization.
for k ∈ {1, 2} do
    Select the valid prognostic region D_k^+ = {(x, y_k, t, s) ∈ D : y_k > 0}.
    for each session s_i do
        Sort D_k^+(s_i) by timestamp and split it into B = 10 consecutive temporal blocks.
        For each block, assign the first 70% to training, the next 10% to validation, and the final 20% to testing.
    end for
    Concatenate the session-wise partitions to obtain D_k^tr, D_k^val, and D_k^te.
    Fit any preprocessing transform only on X_k^tr and apply the fitted transform to X_k^val and X_k^te.
    Train an independent XGBoost regressor
```

$$\hat{f}_k = \arg \min_{f \in \mathcal{F}_{\mathrm{XGB}}} \sum_{(\mathbf{x}, y_k) \in \mathcal{D}_k^{\mathrm{tr}}} \ell(y_k, f(\mathbf{x})) + \Omega(f),$$

```
    using D_k^val for early stopping.
    for S ∈ {tr, val, te} do
        Predict ŷ_k = f̂_k(X_k^S) and compute MAE, RMSE, R^2, the asymmetric PHM score ,PH_med, and PH_cov.
    end for
    Compute SHAP values and PDPs on the held-out test partition D_k^te.
end for
return f̂_1, f̂_2, metrics, and SHAP/PDP results.
```

6.3.2 Model 4: LSTM-Attention for RUL Estimation

The choice of a LSTM network with an attention mechanism follows the recommendation in [61], which highlights its suitability for regression tasks where past events strongly influence the current condition.

Algorithm 9 Multi-Head Self-Attention

Require: Input tensor $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$
Ensure: Context-aware representation $\mathbf{Z} \in \mathbb{R}^{B \times T \times d_{\text{model}}}$
1: **Linear projections**
2: $\mathbf{Q} = \mathbf{X}\mathbf{W}^Q \quad \mathbf{K} = \mathbf{X}\mathbf{W}^K \quad \mathbf{V} = \mathbf{X}\mathbf{W}^V$
3: Reshape into H heads with dimensions d_k, d_v
4: **Scaled dot-product attention**

$$\mathbf{O}_h = \mathrm{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}}\right)\mathbf{V}_h$$

5: **Concatenate heads**

$$\mathbf{O} = \mathrm{Concat}(\mathbf{O}_1, \dots, \mathbf{O}_H)$$

6: **Output projection**

$$\mathbf{Z} = \mathbf{O}\mathbf{W}^O$$

7: **return Z**

The detailed parameter configuration of the LSTM model is provided in Table 20. The mathematical formulation of the attention mechanism is presented in Algorithm 10, while the complete training and evaluation pipeline is outlined in Algorithm 10.

Table 20: LSTM + Multi-Head Attention configuration for RUL prediction.

Parameter	**Value**
Model type	Keras Functional API: LSTM sequence encoder with multi-head self-attention and dense regression head.
Input shape	(W, F), where W = `WINDOW_SIZE` and $F = \lvert \mathbf{X}_{\text{cols}} \rvert$.
LSTM layer	`LSTM` with 128 units, `return_sequences=True`, activation tanh.
Normalization (after LSTM)	`LayerNormalization`.
Dropout (after LSTM)	0.2.
Attention mechanism	`MultiHeadAttention` with $H = 4$ heads, key dimension $d_k = 64$, applied as self-attention on the sequence.
Normalization (after attention)	`LayerNormalization`.
Dropout (after attention)	0.2.
Pooling layer	`GlobalAveragePooling1D` for sequence-level embedding.
Dense head	`Dense`(64, ReLU) $\rightarrow$ `Dropout`(0.1) $\rightarrow$ `Dense`(32, ReLU).
Output layer	`Dense`(1) with `softplus` activation to enforce non-negative RUL predictions.
Optimizer	Adam optimizer with learning rate 1×10^{-3}.
Loss function	Mean Squared Error (MSE).
Evaluation metric	Mean Absolute Error (MAE).
Training epochs	31 epochs.
Batch size	64.
Callbacks	`EarlyStopping` (patience = 10, monitor `val_loss`, restore best weights); `ReduceLROnPlateau` (factor = 0.1, patience = 5, monitor `val_loss`); `EpochPrinter` (log every 200 batches).
Model persistence	Trained model saved as a `.keras` file for each target.

LSTM-Attention: Target-aware temporal split with purge.

For the LSTM-Attention models, a target-aware temporal split with purge intervals was adopted to prevent information leakage caused by overlapping sequence windows. For each RUL target and session, only the valid prognostic region, i.e., $y(t) > 0$, was retained. The samples were sorted chronologically and divided into repeated mission

Algorithm 10 Target-aware LSTM with Multi-Head Self-Attention protocol for state-transition RUL prediction

Require: Cleaned dataset $\mathcal{D} = \{(\mathbf{x}_i, y_{1,i}, y_{2,i}, t_i, s_i)\}_{i=1}^{N}$, with hydraulic features $\mathbf{x}_i \in \mathbb{R}^F$, targets $y_1 = \mathrm{RUL}_{\mathrm{state}}$ and $y_2 = \mathrm{RUL}_{\mathrm{crit}}$, timestamps t_i, and session identifiers s_i
Ensure: Trained models $\hat{g}_1$, $\hat{g}_2$, scalers, metrics, split reports, and SHAP/attention-based visualizations

```
Remove samples with missing values in x, y_1, y_2, t, or s.
Define the sequence length W = 30 and set the purge length P = W.
for k ∈ {1, 2} do
    Select the valid prognostic region
```

$$\mathcal{D}_k^+ = \{(\mathbf{x}, y_k, t, s) \in \mathcal{D} : y_k > 0\}.$$

```
    for each session s_i do
        Sort D_k^+(s_i) chronologically by t.
        Divide the session-wise prognostic sequence into B = 10 consecutive temporal blocks.
        for each block b_j do
            Assign the first 70% of samples to training.
            Discard the following P samples as a purge interval.
            Assign the next 10% of samples to validation.
            Discard the following P samples as a second purge interval.
            Assign the remaining samples to testing.
        end for
    end for
    Concatenate the session- and block-wise partitions into D_k^tr, D_k^val, D_k^te, and discarded purge set D_k^purge.
    Verify that train, validation, and test subsets contain non-empty and non-constant RUL targets.
    Fit a StandardScaler only on X_k^tr and apply the fitted transform to X_k^val and X_k^te.
    Construct LSTM samples {(S_m, y_{k,m}, t_m)} with S_m ∈ R^{W×F} independently within each session, split block, and data partition. The target y_{k,m} is assigned at the final timestep of each window.
    Define an LSTM–MHA regressor
```

$$\hat{y}_k = \hat{g}_k(\mathbf{S}) = \phi_{\mathrm{softplus}}\left(h_\theta\left(\mathrm{MHA}(\mathrm{LSTM}(\mathbf{S}))\right)\right),$$

```
    where the LSTM has 128 units, the multi-head self-attention layer uses H = 4 heads with key dimension d_k = 64, and h_θ denotes the dense regression head.
    Train ĝ_k on (S_k^tr, y_k^tr) using MSE loss, Adam optimization, batch size 64, early stopping on validation loss, and learning-rate reduction on plateau.
    for S ∈ {tr, val, te} do
        Predict ŷ_k^S = max(0, ĝ_k(S_k^S)).
        Sort predictions by the final-window timestamp and compute MAE, RMSE, R^2, the asymmetric PHM score, PH_med, and PH_cov.
    end for
    Generate timestamp-aligned prediction plots with the explicit train–purge–validation–purge–test split map.
    Compute SHAP explanations using training sequences as background and evaluate feature, temporal, and timestep-level importance on train and test sequences.
    Extract multi-head attention scores and compare the averaged temporal attention profile with temporal SHAP importance.
end for
return ĝ_1, ĝ_2, scalers, metrics, split reports, prediction plots, and SHAP/attention visualizations.
```

blocks. Within each block, the first 70% of samples were assigned to training, followed by a purge interval, then 10% for validation, a second purge interval, and the remaining samples for testing.

The purge length was set equal to the LSTM window size. These discarded samples were not used for model training, validation, or testing, but served only to separate adjacent temporal partitions. This avoids sequence overlap across split boundaries and ensures that validation and test windows do not contain temporal context already seen during training. In addition, LSTM sequences were generated independently within each session and split block, so that no input window crossed session, block, or split boundaries. Feature normalization was performed by fitting the scaler only on the training subset and applying the fitted transformation to validation and test data.

This protocol evaluates whether the recurrent model can infer later degradation behavior from earlier temporal patterns under comparable missions, while preserving causal ordering and avoiding leakage through shared sequence history. The split strategy, including the purge intervals, is illustrated in Figure 13 for one representative session.

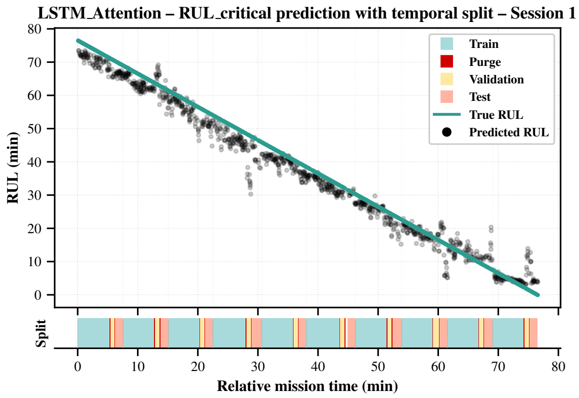


Fig. 13: Example of the target-aware temporal split with purge intervals for the LSTM-Attention model in one representative experimental session. The upper panel compares true and predicted critical RUL, while the lower barcode shows the chronological train, purge, validation, and test partitions. Purge samples are discarded to prevent overlapping LSTM windows from crossing split boundaries.

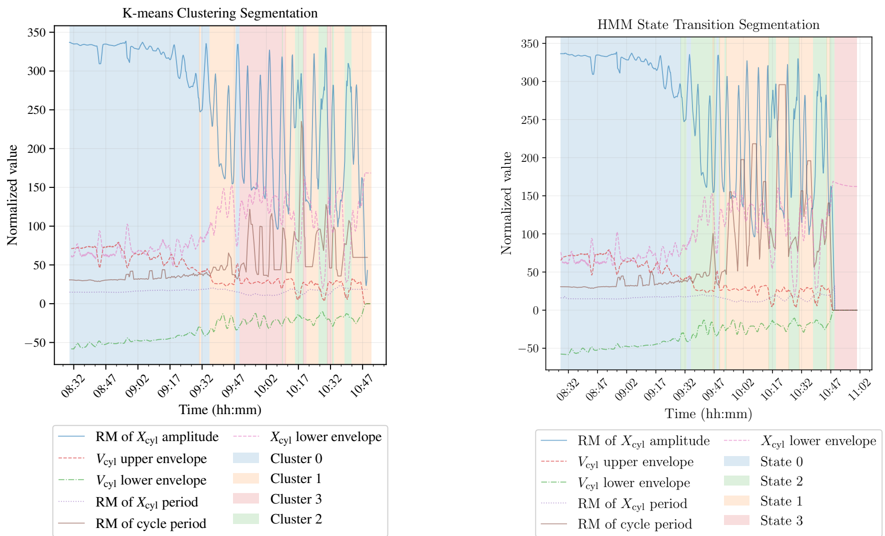
7 Results

7.1 Results Classification: HMM & Kmeans

Detecting anomalous machine conditions requires a clear differentiation of operational states. Traditionally, this relies on expert knowledge; however, unsupervised machine learning methods such as K-Means and Hidden Markov Models (HMM) can support automated clustering of behavior patterns based on derived features [60].

Figure 14a illustrates state segmentation using K-Means on envelope and velocity features, revealing clusters without prior labeling. Conversely, Figure 14b demonstrates the HMM application, which incorporates sequential dependencies and probabilistic transitions. Their comparison in Figures 14 highlights that K-Means yields sharper cluster boundaries, whereas HMM provides smoother state transition modeling.

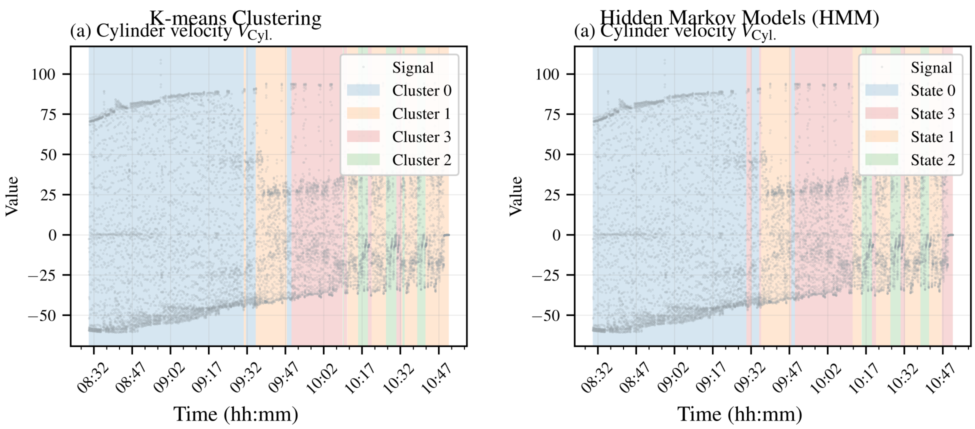
The analysis relies on derived features, where $\mathrm{RM}(\cdot)$ denotes a rolling mean used to smooth short-term fluctuations, and upper/lower envelopes capture the bounding trajectories of X_{cyl} and V_{cyl}.



(a) State segmentation using K-Means clustering on derived features.

(b) State segmentation using Hidden Markov Models (HMM) on the same features.

Fig. 14: Comparison of unsupervised state segmentation approaches applied to hydraulic cylinder signals.



(a) K-means-based cluster assignment for machine state segmentation.

(b) Hidden Markov Model-based state sequence inferred from clustered features.

Fig. 15: Comparison of unsupervised state segmentation methods applied to hydraulic cylinder signals.

For the failure modes *Excessive Internal Leakage* and *Anomalous Operation (Slowdown)*, latent operating states emerging from unsupervised clustering were interpreted and refined using expert knowledge (Table 21). These states are not supervised labels, but semantic representations of cluster-wise statistical patterns grounded in established hydraulic actuator behavior.

Based on this interpretation, the first RUL target is defined as the remaining time until the transition from normal operation (state 0) to abnormal operation (state 1). The critical RUL target is defined as the remaining time until the system reaches a severe operating condition, represented by the transition toward critical or supercritical states (states 3 and 2, respectively). The HMM state sequence is used for RUL target construction because its temporal smoothing reduces short-lived state fluctuations compared with direct K-Means assignments.

Table 21: Description of latent states used in state-based degradation modeling.

Latent state	Interpreted operating condition
0	Normal operation.
1	Abnormal / warning operation.
3	Critical operation.
2	Super-critical operation.

7.1.1 PLC-Referenced Evaluation Protocol for State Classification

The unsupervised state models were evaluated against PLC-generated operational event logs, which were used as an external reference layer rather than as physical ground-truth fault labels. The PLC logs were not used for training K-Means or HMM. They were only used after model inference to quantify the agreement between data-driven latent states and independently generated operational condition intervals.

For each timestamp t_i, a PLC reference label $y_{\mathrm{PLC}}(t_i)$ was assigned from the logged event intervals:

$$y_{\mathrm{PLC}}(t_i) = \begin{cases} c_j, & t_i \in [t_{\mathrm{start},j}, t_{\mathrm{end},j}], \\ 1, & \text{otherwise}, \end{cases}$$

where $c_j \in \{2, 3\}$ denotes warning or critical operation. Thus, the PLC reference scale is defined as normal operation (1), warning/abnormal operation (2), and critical operation (3).

Because K-Means and HMM are unsupervised models, their raw state indices have no intrinsic semantic meaning. A deterministic state-to-condition mapping was therefore defined before computing the evaluation metrics. The mapping was based on the hydraulic signatures of each state, including velocity-envelope behavior, cycle-period variation, pressure-ratio behavior, and expert interpretation of the operating regime. Since the PLC taxonomy does not distinguish between critical and supercritical degradation, both severe latent states were mapped to the PLC critical level (Table 22).

Table 22: Mapping from unsupervised latent states to PLC operational reference levels.

Raw state	Interpreted state	Hydraulic interpretation	PLC level
0	Normal	Stable and repeatable cyclic response	1
1	Warning / abnormal	Degraded but still operational response	2
3	Critical	Severe degradation or unstable operating regime	3
2	Super-critical	Near non-viable operation or highly degraded regime	3

The mapped predictions were obtained as

$$\hat{y}_{\mathrm{KM}}(t_i) = \pi_{\mathrm{KM}}\left(s_{\mathrm{KM}}(t_i)\right), \qquad \hat{y}_{\mathrm{HMM}}(t_i) = \pi_{\mathrm{HMM}}\left(s_{\mathrm{HMM}}(t_i)\right),$$

where $\pi(\cdot)$ denotes the mapping in Table 22. The mapped predictions were then compared with $y_{\mathrm{PLC}}(t_i)$ using accuracy, macro-averaged precision, recall, and F1-score, weighted F1-score, and binary anomaly F1-score. For the binary metric, warning and critical samples were grouped into a single non-normal class:

$$y_{\mathrm{bin}}(t_i) = \begin{cases} 0, & y_{\mathrm{PLC}}(t_i) = 1, \\ 1, & y_{\mathrm{PLC}}(t_i) \in \{2, 3\}. \end{cases}$$

This protocol quantifies agreement with PLC-level operational monitoring, not absolute physical fault-label accuracy. This distinction is important because PLC events may include threshold hysteresis, delayed alarm activation, conservative alarm settings, or operator-defined contextual annotations. Therefore, the reported metrics should be interpreted as operational agreement between the unsupervised state models and the industrial reference layer.

Figure 16 visualizes the evaluation procedure. The top panel shows the cylinder velocity signal with PLC event intervals, followed by the PLC reference track, the mapped K-Means and HMM predictions, and the corresponding raw latent states. This representation separates the original unsupervised state space from the mapped PLC-compatible operational scale used for quantitative evaluation.

Table 23: Agreement between PLC operational reference labels and mapped K-Means/HMM state predictions.

Model	Acc.	Macro Prec.	Macro Rec.	Macro F1	Weighted F1	Binary F1
K-Means	0.737	0.689	0.785	0.624	0.789	0.938
HMM	0.749	0.665	0.790	0.635	0.790	0.933

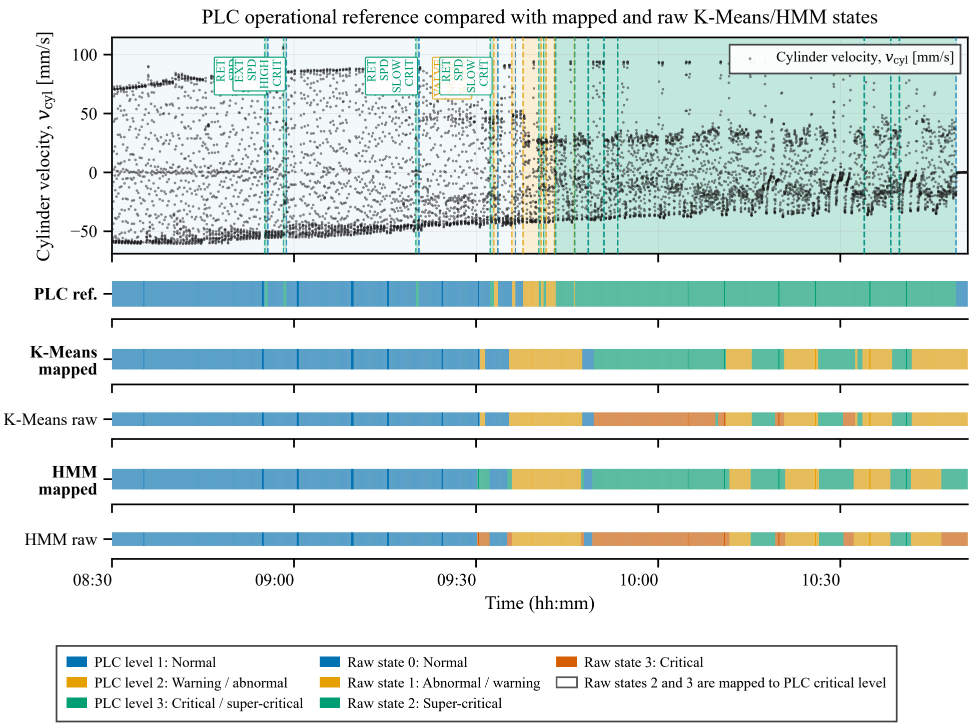


Fig. 16: PLC-referenced evaluation of unsupervised state classification. The raw K-Means and HMM states are shown separately from the mapped operational predictions to make the state-to-condition mapping transparent. Critical and super-critical latent states are both mapped to the PLC critical level because the PLC reference taxonomy contains only one critical class.

The results show that both unsupervised models achieve comparable agreement with the PLC operational reference. The HMM provides a slightly higher overall accuracy and macro F1-score, indicating more balanced performance across the three operating levels. K-Means achieves a slightly higher binary anomaly F1-score, suggesting strong separation between normal and non-normal operation. The difference between macro F1 and weighted F1 reflects the class imbalance of the PLC reference, especially the lower support of the warning class. Therefore, the binary anomaly F1-score is reported as an additional operational metric, since industrial monitoring is often primarily concerned with detecting deviations from normal operation.

7.2 Results Model 3: XGBoost Regressor

Table 25 summarizes the model performance, including a PHM-based metric adapted for minute-level RUL evaluation of the hydraulic system, computed following Algorithm 11 and the formulation in [62].

Table 24: Confusion matrices for mapped K-Means and HMM predictions against the PLC operational reference. Rows denote PLC reference labels and columns denote predicted labels.

Model	PLC reference	Pred. normal	Pred. warning	Pred. critical
K-Means	Normal	18010	1352	3
	Warning	204	1544	0
	Critical	1194	8167	11032
HMM	Normal	17640	382	1343
	Warning	204	1530	14
	Critical	1063	7413	11917

Algorithm 11 PHM score computation.

Require: Ground truth $\mathbf{y}_{\text{true}} = (y_{\text{true},1}, \ldots, y_{\text{true},N})$, predictions $\mathbf{y}_{\text{pred}} = (y_{\text{pred},1}, \ldots, y_{\text{pred},N})$
Require: Parameters Δ^{+}_{crit}, Δ^{-}_{tol}, penalty levels P_{+}, P_{-}
Ensure: PHM_{tot} and PHM_{avg}

1: Compute penalty scales

$$\beta \leftarrow \frac{\Delta^{+}_{\text{crit}}}{\ln(1+P_{+})}, \qquad \alpha \leftarrow \frac{\Delta^{-}_{\text{tol}}}{\ln(1+P_{-})}.$$

2: Compute residuals for $i = 1, \ldots, N$

$$d_i \leftarrow y_{\text{pred},i} - y_{\text{true},i}.$$

3: Compute asymmetric cost for each residual

$$s_i \leftarrow \begin{cases} \exp(-d_i/\alpha) - 1, & d_i < 0 \quad \text{(conservative)} \\ \exp(d_i/\beta) - 1, & d_i \geq 0 \quad \text{(optimistic).} \end{cases}$$

4: Aggregate PHM scores

$$\text{PHM}_{\text{tot}} \leftarrow \sum_{i=1}^{N} s_i, \qquad \text{PHM}_{\text{avg}} \leftarrow \frac{\text{PHM}_{\text{tot}}}{N}.$$

Note: Lower values indicate better performance. Optimistic errors ($d_i \geq 0$) incur higher penalties than conservative errors ($d_i < 0$). PHM denotes **Prognostics and Health Management**.

5: **return** PHM_{tot}, PHM_{avg}

Table 25: Performance metrics for RUL prediction using XGBoost models.

Model	Target	Set	MAE	RMSE	R^2	PHM_{avg}	PH_{med}	PH_{cov}
XGB	RUL_1	Train	3.164	3.991	0.955	0.070	39.474	0.873
XGB	RUL_1	Val	3.913	5.193	0.924	0.091	34.035	0.780
XGB	RUL_1	Test	3.916	5.625	0.911	0.092	34.068	0.847
XGB	RUL_c	Train	3.444	4.573	0.958	0.077	45.903	0.887
XGB	RUL_c	Val	5.612	7.567	0.883	0.133	41.023	0.787
XGB	RUL_c	Test	4.981	6.268	0.920	0.115	47.347	0.740

Note: XGB denotes XGBoost. RUL_1 denotes the state-transition RUL, while RUL_c denotes the critical RUL. PHM scores follow the asymmetric prognostics penalty formulation, where lower values indicate better performance. PH_{med} is reported in minutes. PH_{cov} denotes the fraction of session/block trajectories for which a valid prognostic horizon was identified.

Feature Explainability and Selection

Feature explainability was assessed using SHAP values [63] for local feature contributions and Partial Dependence Plots (PDPs) [64] for global marginal effects. Figures 18 and 17 show the SHAP dependence plots and the aggregated SHAP summary for the XGBoost model trained for *RUL* prediction, while Figures 19 report the corresponding PDPs.

The SHAP summary identifies the previous-cycle minimum cylinder velocity, previous hydraulic cycle duration, pressure ratio $P_{B/A}$, and oil flow rate Q_{oil} as the dominant predictors. This ranking is consistent with the expected hydraulic behavior in the investigated leakage-oriented degradation scenario,

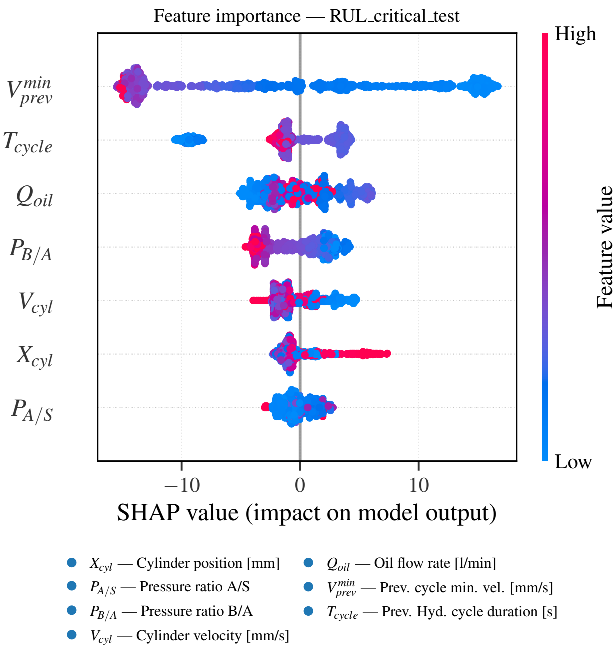


Fig. 17: SHAP summary plot showing global feature importance for RUL prediction using XGBoost.

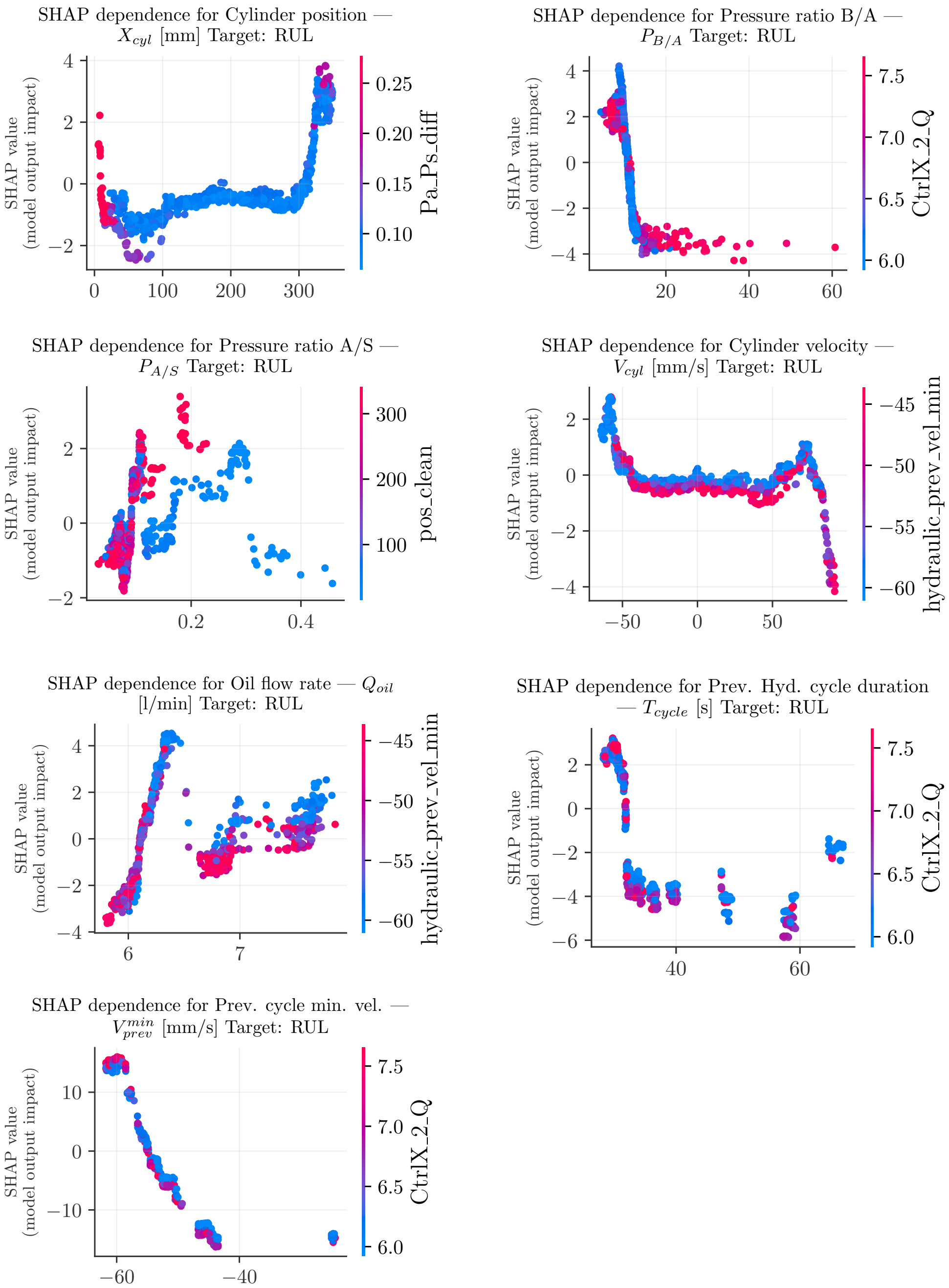


Fig. 18: SHAP dependence plots showing the local influence of the selected hydraulic and cycle-level features on XGBoost-based RUL prediction.

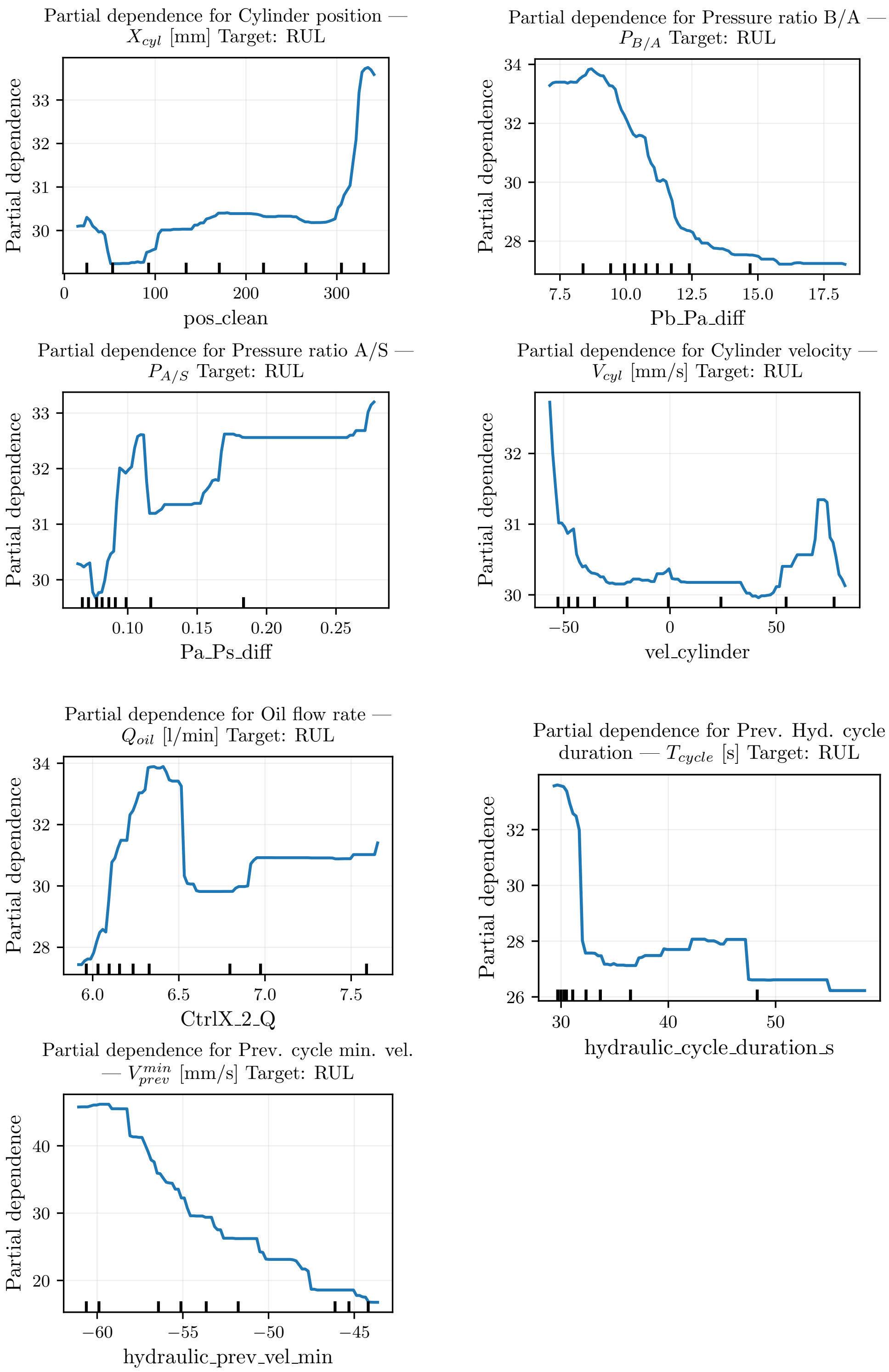


Fig. 19: Partial Dependence Plots showing the marginal effect of the selected hydraulic and cycle-level features on XGBoost-based RUL prediction.

7.3 Results model 4: LSTM-Attention

To evaluate the impact of the attention mechanism on convergence and model performance, we conducted 25 independent training runs (10 epochs each) using the architecture detailed in Table 20. For comparison, the attention layer was replaced only in the evaluation scenario by a global average pooling operation (`x = tf.keras.layers.GlobalAveragePooling1D()(x)`), while keeping all other hyperparameters and training settings identical. The averaged results are reported in Table 26.

Table 26: Impact of the attention mechanism on LSTM-based RUL prediction (average over 25 runs, 10 epochs per run).

Model	Target	Set	MAE	RMSE	R^2	PHM$_{\text{tot}}$	PHM$_{\text{avg}}$
LSTM	RUL	Train	6.518	11.063	0.661	5086.928	0.167
LSTM	RUL	Val	6.361	10.830	0.667	702.021	0.162
LSTM	RUL	Test	6.576	11.212	0.660	1465.520	0.169
LSTM+Attn	RUL	Train	**2.468**	**5.554**	**0.915**	**1785.685**	**0.059**
LSTM+Attn	RUL	Val	**2.502**	**5.681**	**0.908**	**258.529**	**0.060**
LSTM+Attn	RUL	Test	**2.558**	**5.761**	**0.910**	**533.447**	**0.061**
LSTM	RUL$_c$	Train	7.690	12.685	0.750	6324.223	0.208
LSTM	RUL$_c$	Val	7.548	12.364	0.758	873.121	0.202
LSTM	RUL$_c$	Test	7.749	12.874	0.747	1825.361	0.210
LSTM+Attn	RUL$_c$	Train	**3.882**	**7.539**	**0.912**	**2877.344**	**0.094**
LSTM+Attn	RUL$_c$	Val	**3.863**	**7.541**	**0.910**	**408.075**	**0.094**
LSTM+Attn	RUL$_c$	Test	**3.916**	**7.649**	**0.911**	**835.622**	**0.096**

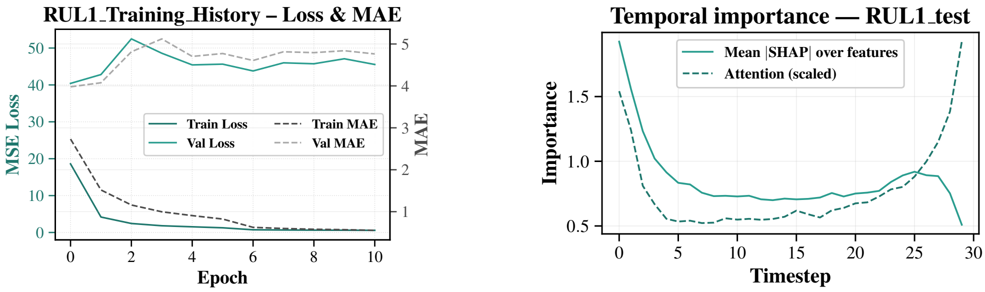
Note: LSTM+Attn denotes an LSTM model with additive attention.

Final performance metrics after 31 training epochs with attention are presented in Table 27.Training history for loss and MAE (Fig. 20), global feature importance (Fig. 22), spatio-temporal attention heatmap (Fig. 21).

Table 27: Performance metrics for RUL prediction using the LSTM with attention mechanism.

Model	Target	Set	MAE	RMSE	R^2	PHM$_{\text{avg}}$	PH$_{\text{med}}$	PH$_{\text{cov}}$
LSTM+Attn	RUL$_1$	Train	2.407	2.977	0.975	0.048	35.912	1.000
LSTM+Attn	RUL$_1$	Val	3.979	6.357	0.886	0.087	33.840	0.793
LSTM+Attn	RUL$_1$	Test	4.194	6.162	0.893	0.092	33.675	0.900
LSTM+Attn	RUL$_c$	Train	2.110	2.692	0.985	0.043	42.006	1.000
LSTM+Attn	RUL$_c$	Val	3.864	5.152	0.946	0.085	32.982	0.853
LSTM+Attn	RUL$_c$	Test	3.467	4.218	0.964	0.075	39.662	0.893

Note: RUL$_1$ denotes the state-transition RUL, while RUL$_c$ denotes the critical RUL. PHM scores follow the asymmetric prognostics penalty formulation, where lower values indicate better performance. PH$_{\text{med}}$ is reported in minutes. PH$_{\text{cov}}$ denotes the fraction of session/block trajectories for which a valid prognostic horizon was identified.



(a) Training loss and MAE evolution

(b) Temporal SHAP vs. attention weights

Fig. 20: (a) Training loss progression across 31 epochs for the LSTM + Attention model, showing convergence and stability of both loss and MAE metrics. (b) Comparison between temporal SHAP importance values and normalized multi-head attention weights for RUL_1 prediction on the test set, showing a similar temporal pattern between SHAP-based temporal importance and attention weights.

Feature Explainability and Selection

In Fig. 21, a consistent attention pattern is visualized, obtained by extracting the weights from the MultiHeadAttention layer. The function builds a submodel to retrieve attention scores, averages them across heads and samples, and computes the per-timestep attention vector used in the heatmap.

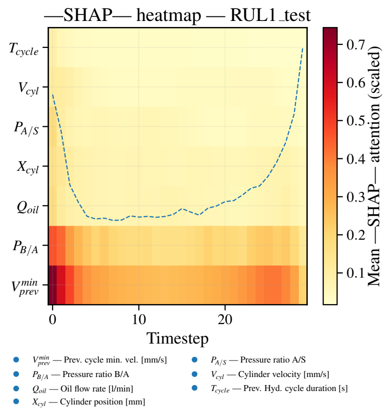


Fig. 21: Spatio-temporal attention heatmap across features and timesteps. Higher attention weights indicate stronger model focus within the learned attention mechanism and are interpreted only as auxiliary indicators, not as causal explanations.

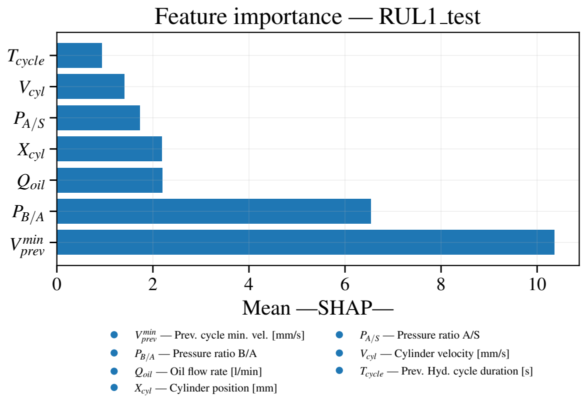


Fig. 22: SHAP-based global feature importance on the test set for the LSTM + Attention model.

7.4 Summary of Experimental Findings

The experimental results show that the proposed state-based framework provides a coherent prototype-level basis for hydraulic predictive maintenance under the investigated operating conditions. For unsupervised state identification, K-Means and HMMs exhibited complementary behavior. K-Means produced sharper geometric partitions in the engineered feature space and achieved the highest binary anomaly F1-score (0.938), indicating strong separation between PLC-referenced normal and non-normal intervals. In contrast, the HMM achieved slightly higher overall accuracy (0.749) and macro F1-score (0.635), reflecting more balanced agreement across the PLC-referenced operating levels. This supports the use of K-Means as a simple state-separation baseline, while the HMM provides a more temporally coherent basis for constructing state-transition RUL targets.

For state-transition RUL estimation, the nonlinear models outperformed the linear baselines under the same target-aware temporal test protocol (Table 28). For RUL_1, linear and ridge regression both achieved a test MAE of 6.878 min and $R^2 = 0.714$, whereas XGBoost reduced the MAE to 3.916 min and increased R^2 to 0.911. The LSTM-Attention model achieved a comparable test MAE of 4.194 min and $R^2 = 0.893$. The prognostic horizon metrics further indicate that the linear baselines did not provide a useful early horizon for RUL_1, while XGBoost and LSTM-Attention reached median horizons of 34.068 min and 33.675 min, respectively. This suggests that the first transition into abnormal operation is strongly represented by the engineered hydraulic and cycle-level features, which can be effectively captured by a tabular nonlinear model.

For the critical RUL target, sequence modeling showed a clearer advantage. Linear and ridge regression both reached a test MAE of 9.494 min and $R^2 = 0.702$, while XGBoost reduced the MAE to 4.981 min and achieved $R^2 = 0.920$. The LSTM-Attention model achieved the best critical RUL performance, with a test MAE of 3.467 min, RMSE of 4.218 min, $R^2 = 0.964$, and the lowest average PHM penalty. Although XGBoost obtained the largest median prognostic horizon for RUL_c

(47.347 min), its lower PH coverage indicates less consistent valid horizons across evaluated trajectories. In contrast, the LSTM-Attention model combined lower prediction error with higher PH coverage, suggesting more stable critical RUL prediction. Overall, these results indicate that the progression toward critical operation depends more strongly on accumulated temporal context than the first abnormal transition.

Table 28: Test-set RUL prediction performance.

Model	Target	MAE	RMSE	R^2	$\mathrm{PHM}_{\mathrm{avg}}$	$\mathrm{PH}_{\mathrm{med}}$	$\mathrm{PH}_{\mathrm{cov}}$
Linear Regression	RUL_1	6.878	10.061	0.714	0.174	0.000	0.000
Ridge Regression	RUL_1	6.878	10.061	0.714	0.174	0.000	0.000
XGBoost	RUL_1	**3.916**	**5.625**	**0.911**	**0.092**	**34.068**	0.847
LSTM+Attn	RUL_1	4.194	6.162	0.893	**0.092**	33.675	0.900
Linear Regression	RUL_c	9.494	12.092	0.702	0.247	0.429	1.000
Ridge Regression	RUL_c	9.494	12.092	0.702	0.247	0.429	1.000
XGBoost	RUL_c	4.981	6.268	0.920	0.115	**47.347**	0.740
LSTM+Attn	RUL_c	**3.467**	**4.218**	**0.964**	**0.075**	39.662	0.893

Note: RUL_1 denotes the state-transition RUL, while RUL_c denotes the critical RUL. $\mathrm{PHM}_{\mathrm{avg}}$ is the average asymmetric prognostics penalty; lower values indicate better performance. $\mathrm{PH}_{\mathrm{med}}$ is the median prognostic horizon in minutes, and $\mathrm{PH}_{\mathrm{cov}}$ is the fraction of evaluated trajectories with a valid prognostic horizon. A high $\mathrm{PH}_{\mathrm{cov}}$ with a near-zero $\mathrm{PH}_{\mathrm{med}}$ indicates that the tolerance criterion is satisfied only immediately before the transition.

8 Discussion

The results show that the proposed architecture provides a prototype-level basis for state-based predictive maintenance in the investigated hydraulic system by connecting PLC-based operational logs, high-frequency hydraulic measurements, latent state modeling, and state-transition RUL estimation. The main implication is not that a single model provides universally transferable prognostics, but that an operationally traceable data architecture can provide a practical basis for RUL estimation when run-to-failure trajectories and component-level failure labels are unavailable.

It should be noted that the supervised RUL targets are constructed from the inferred HMM state sequence and therefore represent model-defined state-transition targets. The PLC-based event logs are not used to generate the HMM states, but are used as an independent operational reference to assess whether the inferred transitions are consistent with recorded non-nominal machine behavior. Therefore, the evaluation supports operational agreement rather than fully independent physical ground-truth validation.

8.1 Interpretability and Observability

The explainability analyses indicate that the RUL models relied mainly on hydraulically plausible variables, including cylinder velocity, pressure ratios, oil flow, and cycle-level descriptors. This is consistent with the investigated leakage and slowdown behavior, where pressure imbalance, altered flow demand, and reduced cycle repeatability are expected to precede non-nominal operation.

However, SHAP values, PDPs, and attention maps are interpreted here only as evidence of physical consistency, not as proof of physical causality. In hydraulic systems, measured variables are strongly coupled, and unmeasured effects such as temperature, viscosity, local leakage paths, valve behavior, or component efficiency may influence the observed signals. Therefore, the proposed framework supports a system-level prognostic interpretation under the available sensing configuration, but it does not claim component-level causal diagnosis.

This reinforces the architectural role of the proposed event and data backbone. Reliable interpretation depends not only on the learning algorithm, but also on synchronized sensor data, structured PLC-based event logs, and physically meaningful feature construction. The operational reference layer is therefore central in this study for relating latent states and RUL targets to observable machine behavior.

8.2 Scope and Limitations

The framework was evaluated on a real PLC-controlled electro-hydraulic prototype under repeated and controlled operating conditions. Although the hardware, control logic, and acquisition stack are industrially representative, the study does not yet demonstrate transferability across multiple hydraulic installations, broader operating regimes, or long-term production environments.

The proposed RUL definition should also be interpreted within this scope. Because complete run-to-failure trajectories and verified component-level failure records were not available, RUL was defined as the time or number of cycles before transition to a PLC-supported abnormal or critical operating state. This provides a planning-oriented prognostic target for the investigated setting, but it is not intended to replace component-level lifetime models when complete degradation and replacement histories are available.

A direct economic validation was also outside the scope of this prototype. The event-logging layer records the timing, duration, and category of non-nominal states, and the RUL pipeline estimates the remaining time before abnormal or critical operation. These elements provide a basis for future availability- or OEE-oriented analysis, but this study does not quantify cost savings, return on investment, scrap reduction, or production-line downtime reduction.

8.3 Operational Relevance and Economic Interpretation

The proposed architecture is operationally relevant because it links non-nominal machine states with their timing, duration, and predicted transition horizons. This information can support future availability- and OEE-oriented analyses by relating state-transition RUL estimates to downtime and maintenance records. However, the present prototype was evaluated in a controlled commissioning environment and does not include production-line variables such as throughput, scrap, downtime cost, maintenance cost, or recovery time. Therefore, this study does not quantify cost savings, return on investment, or direct economic impact; such validation would require deployment in a production environment with synchronized operational, maintenance, and economic records.

8.4 Outlook: Toward Physics-Informed Health-Indicator RUL

The state-transition RUL formulation adopted in this work is appropriate for the available observability level, where high-frequency hydraulic measurements are combined with PLC-based operational event logs. When richer component-level measurements become available, the same architecture could be extended toward a physics-informed health-indicator RUL formulation.

In such an extension, degradation could be represented through continuous health indicators derived from hydraulic efficiency, leakage-related pressure–flow consistency, temperature-viscosity compensation, actuator response delay, or cycle-level repeatability. Instead of defining RUL only as the transition between latent operational states, RUL could then be defined as the first time at which a physics-informed health indicator crosses a critical threshold:

$$T_{\mathrm{HI}} = \inf\{t > t_k : HI(t) \leq HI_{\mathrm{crit}}\}, \qquad RUL_{\mathrm{HI}}(t_k) = T_{\mathrm{HI}} - t_k. \tag{35}$$

For cyclic hydraulic systems, the same idea can be expressed in terms of the remaining number of extension–retraction cycles before the health indicator reaches the critical threshold.

Graph-based modeling provides a possible way to structure such health indicators when sufficient sensing is available. In this case, measured variables, operating-context information, degradation proxies, and health indicators could be represented as nodes, while edges encode known or hypothesized hydraulic relationships. However, such a graph should not be learned only from statistical correlation if it is intended to support physical interpretation. To be useful as an explainable prognostic layer, the graph structure should be constrained by hydraulic topology, conservation relationships, operating regimes, and expert knowledge.

This extension is not implemented as the primary RUL formulation in the present study because it requires higher component-level observability than is available in the current prototype. It is therefore considered a future research direction for more highly instrumented Fluid 4.0 deployments, where physics-informed graph structures may complement state-transition RUL by supporting more explicit links between measured variables, degradation mechanisms, and maintenance decisions.

9 Conclusion

This work presented an edge-oriented prototype architecture for state-based predictive maintenance in hydraulic and fluid power systems. The contribution is primarily architectural and methodological: the proposed framework connects OPC UA-based hydraulic data acquisition, PLC-based operational event logs, unsupervised state modeling, state-transition RUL estimation, and explainability analysis within a single prototype-level workflow. The study addresses practical hydraulic PdM constraints, including limited failure labels, restricted component-level observability, and the absence of complete run-to-failure data.

The main conclusions are aligned with the contributions of this work:

- **Edge-oriented data and event architecture.** The proposed acquisition and ETL architecture aligned high-frequency hydraulic sensor data, multi-resolution storage, and PLC-based event logs. In the investigated prototype, the 100 ms data layer preserved short-term pressure–flow–motion dynamics, while the event logs provided the operational reference needed to interpret non-nominal machine states.
- **State-based condition modeling under limited labels.** K-Means and HMMs were used to infer latent operating states from unlabeled hydraulic time-series data. These states were interpreted using hydraulic signal behavior, PLC event intervals, and expert knowledge, but not treated as physical ground-truth failure labels. K-Means achieved the highest binary anomaly F1-score of 0.938, while the HMM achieved slightly higher overall agreement with the PLC reference layer, with an accuracy of 0.749 and a macro F1-score of 0.635.
- **State-transition RUL formulation and evaluation.** RUL was defined as the remaining time, or number of hydraulic cycles, before the system transitions from nominal operation to a PLC-supported abnormal or critical operating state. This definition provides an operational prognostic target for hydraulic systems without complete run-to-failure trajectories. In the experimental evaluation, XGBoost achieved the best test MAE for the first abnormal transition, with 3.916 min for RUL_1, while LSTM-Attention achieved the best critical RUL performance, with 3.467 min MAE and $R^2 = 0.964$.
- **Explainability as physical consistency analysis.** SHAP values, PDPs, and attention-based analyses indicated that the models relied mainly on hydraulically plausible variables, including cylinder velocity, previous-cycle minimum velocity, cycle duration, pressure ratios, and oil flow rate. These findings support consistency with the investigated leakage and slowdown behavior, but they are not interpreted as proof of causal degradation mechanisms.

Overall, the results provide prototype-level evidence that the proposed architecture can support operational state identification and state-transition RUL estimation for the investigated hydraulic system. The findings should be interpreted as system-level prognostics under controlled operating conditions, not as proof of cross-system transferability, production-scale scalability, or component-level causal diagnosis.

Future work should evaluate the framework on additional hydraulic systems, longer operating periods, and broader operating regimes. Further extensions should include verified maintenance records, component-level instrumentation, availability- and cost-oriented evaluation, and physics-informed health indicators. Under richer observability, graph-based or physics-informed models could be incorporated to represent hydraulic topology more explicitly and strengthen the link between measured variables, degradation mechanisms, and maintenance decisions.

Appendix A Experimental Testbed: Hydraulic Unit

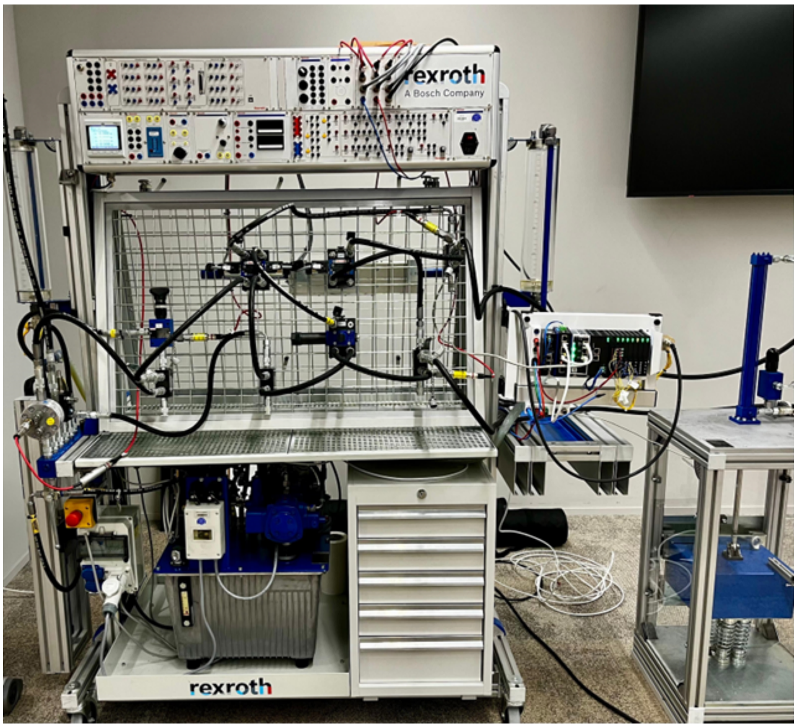


Fig. A1: Schematic of the hydraulic unit used for data acquisition and model evaluation.[56]

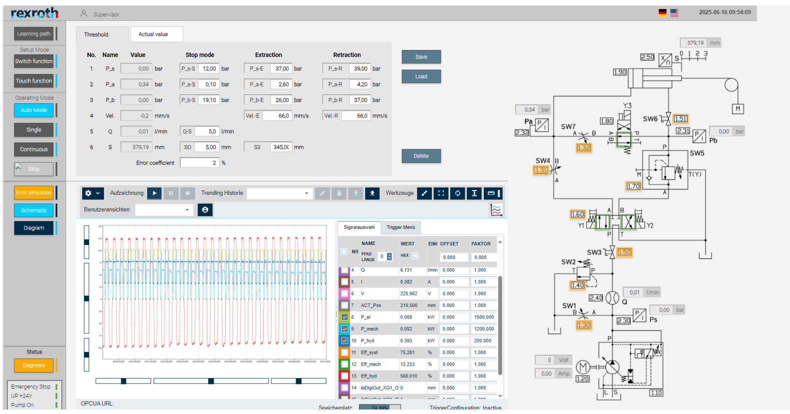


Fig. A2: Operator GUI for visualizing and validating PLC signal activity.[56]

Fig. A3: Visualization of model prediction results for continuous evaluation on test data.

Appendix B Data Flow - Orchestration

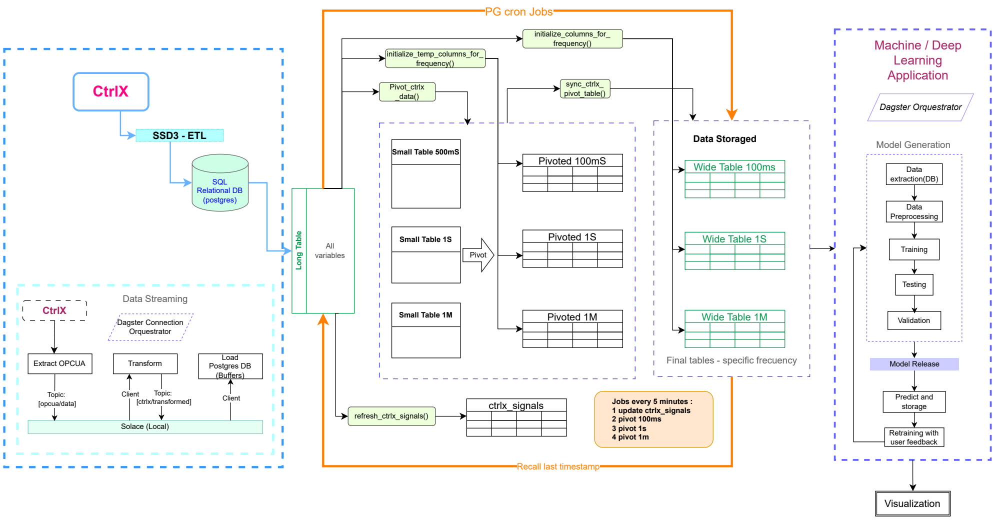


Fig. B4: ETL data flow from PLCs to time-series storage for real-time sensor monitoring and predictive analysis.

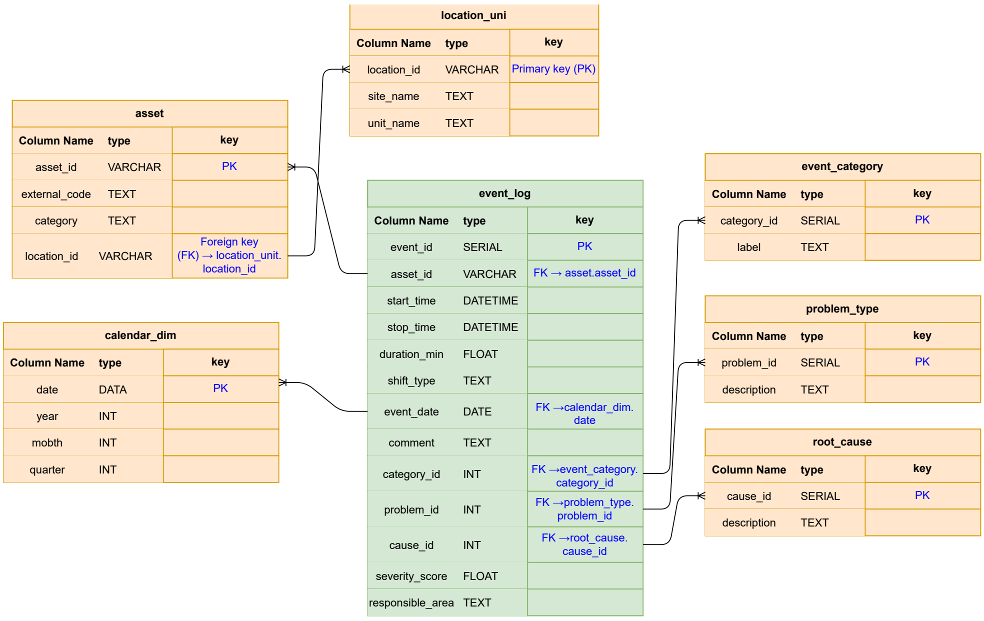


Fig. B5: Normalized relational schema for logging machine faults and operational events as the foundation for predictive maintenance.

Appendix C PLC Event-Code Taxonomy and Event-Log Structure

This appendix reports the operational encoding used to generate the PLC-based reference labels. The taxonomy converts rule-based PLC events and manual contextual annotations into structured operational reference records. These records are not treated as physical ground truth, but as an industrially motivated reference layer for state comparison, RUL-target construction, and operational traceability.

Table C1 summarizes the condition levels used in the PLC event logs. Table C2 reports the event-code taxonomy used in the representative degradation session, while Table C3 shows the corresponding structure of representative event-log records.

Table C1: Encoding of operational condition and severity levels used in the PLC event logs.

Level	Code meaning	Interpretation
1	Normal	Expected hydraulic operation or non-critical standstill condition
2	Warning	Deviation from the expected operating range
3	Critical	Severe abnormal condition requiring maintenance attention

Table C2: PLC event-code taxonomy used for the representative hydraulic session.

Event code	Event label	Cat.	Prob.	Cause	Cond.	Source	Operational interpretation
NORMAL OPERATION	Normal hydraulic operation	1	0	0	1	system condition	No abnormal hydraulic condition was detected.
EXT SPD HIGH WARN	Extension speed above limit	3	102	502	2	hydraulic prev vel max; system condition	Extension velocity exceeded the expected operational threshold.
EXT SPD HIGH CRIT	Extension speed critically high	3	102	512	3	hydraulic prev vel max; system condition	Extension velocity reached a critical abnormal value.
RET SPD SLOW WARN	Retraction speed slower than expected	3	106	506	2	hydraulic prev vel min; system condition	Retraction velocity became slower than expected.
RET SPD SLOW CRIT	Retraction speed critically slow	3	106	516	3	hydraulic prev vel min; system condition	Retraction velocity reached a critically slow condition.
VALVE NOISE WARN	Excessive valve noise	2	201	601	2	manual report	Manual annotation of excessive noise in the directional control valve.
STANDSTILL	Cylinder standstill detected	3	0	0	1	vel cylinder; system condition	Actuator velocity remained within the standstill range.
CYL LEAK STOP CRIT	Cylinder stop due to leakage	3	101	501	3	manual report	Manual critical annotation of excessive leakage and insufficient pressure.

Table C3: Representative structure of the hydraulic event-log records.

ID	Asset	Start	Stop	Dur.	Shift	Date	Comment	Cat.	Prob.	Cause	Sev.	RA
1	AST001	10:10	10:40	30	Day	25-07-30	Cylinder not extending/retracting; internal pressure drop.	3	101	501	7.5	H
2	AST001	13:20	13:50	30	Day	25-07-31	Extension speed unusually slow; likely pump inefficiency.	3	102	502	6.8	M
3	AST003	15:00	15:10	10	Night	25-07-29	Routine inspection of oil cooler; no anomalies found.	2	0	0	2.0	O
4	AST002	08:00	09:15	75	Day	25-08-01	Pressure accumulator failed to hold charge.	3	103	503	8.2	H
5	AST004	22:00	22:45	45	Night	25-08-02	Leak detected at main hose connection.	3	104	504	7.0	M
6	AST001	11:30	12:00	30	Day	25-08-03	Filter blockage indicated by high differential pressure.	3	105	505	6.5	O
7	AST005	06:00	06:15	15	Night	25-08-04	Sensor calibration required.	2	0	0	3.0	I

Note: Duration is given in minutes. Severity score ranges from 1 (minor) to 10 (critical). RA denotes the responsible area: H—Hydraulics, M—Maintenance, O—Operations, I—Instrumentation.

Declarations

Ethical approval

Not applicable.

Consent to Participate

Not applicable.

Consent to Publish

Not applicable.

Data Availability

The datasets generated and/or analysed during the current study are not publicly available due to confidentiality agreements with industrial project partners, but are available from the corresponding author on reasonable request.

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